# LARGE ERDŐS–STRAUS WITNESSES: THE MULTIPLIER PARAMETER EXCEEDS EVERY POWER OF $\log p$

ANONYMOUS

ABSTRACT. For a prime $p$ let $W(p)$ be the least modulus $M \equiv 3 \pmod 4$ of a "multiplier" congruence that forces a solution of $4/p = 1/x+1/y+1/z$, and let $ck_{\min}(p)$ be the least product $ck$ of a positive Type I slice. Both are expected to be at most a power of $\log p$ for almost all $p$, and are known to exceed a constant times $\log p$ for infinitely many $p$. We prove that for every fixed $k$, $W(p) \geq (\log p)^k \exp(-C_k \log\log p/\log\log\log p)$ for infinitely many primes $p$ in Mordell's hard classes; more precisely, the least hard prime with $W(p) > T$ is at most $\exp(T^{o(1)})$ for all large $T$. The class of one is imposed modulo the primes up to $T^{1/k+o(1)}$, so that the residual congruence system consists of events on at most $k-1$ free primes. We build a pointwise minorant of its void indicator by inclusion–exclusion truncated by the number of primes in the support, not by the number of events, and we organise the events in *levels* by support size. Each level is sieved inside the levels below it; by a conditional form of the local lemma its truncation, and hence the codegree threshold it needs, depends only on its own mass. Heavy vertex sets ("hubs", which do occur) are moved to lower levels by a first-moment argument, and the bottom level, a graph, needs no codegree condition at all. The minorant is transferred to primes by the Linnik-range prime number theorem of Thorner and Zaman, whose relative error term absorbs a possible Siegel zero. As a warm-up we give the simpler prime-local construction, which reaches exponent 2 and cannot go further. For the slice parameter we prove that congruence methods certify exactly the least quadratic non-residue, giving $ck_{\min}(p) \gg \log p \log\log\log p$ infinitely often. This is Graham and Ringrose's $\Omega$-bound for the least non-residue, transported to $ck_{\min}$ (in Lau–Wu's form). On the Haar side we prove $\log(1/\delta^*(T)) \leq T^{o(1)}$ for the avoider density (polylogarithmic modulo Elsholtz–Tao), identify the single-prime atoms with points of Elsholtz–Tao's Type I variety, and show that an individual Type I bound $n^\eta$ would give the Haar exponent $\eta/(1+\eta)$.

## 1. INTRODUCTION

The Erdős–Straus conjecture asserts that $4/n = 1/x+1/y+1/z$ is solvable in positive integers for every $n \geq 2$. It suffices to treat primes, it has been verified for $n \leq 10^{18}$ ([11]; [14] for $10^{17}$; see [13], §6), and Mordell's identities settle every prime outside the six classes $p \equiv 1^2, 11^2, 13^2, 17^2, 19^2, 23^2$

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(mod 840) [10] (cf. also [12]); we call such primes *hard*. On the exceptional-set side, Vaughan [16] proved that the number $E(N)$ of $n \le N$ without a representation satisfies $E(N) \ll N \exp\{-c(\log N)^{2/3}\}$; see [13] for a uniform modern account and for many exceptions to the analogous statement with large numerators, and [4] for the average number of solutions, which over primes $p \le N$ is $(\log N)^3$ up to a factor $\log\log N$.

This note is about the *pointwise size* of the smallest witness. Two natural witness parameters are the following.

*The multiplier parameter.* For $M \equiv 3 \pmod 4$ put $A_M = (M+1)/4$. If $uvw = A_M$ and $n \equiv -uv^{-1} \pmod M$, then $s = (nv+u)/M$ is a positive integer and

$$\frac{4}{n} = \frac{1}{suw} + \frac{1}{nsvw} + \frac{1}{nuvw} \tag{1}$$

(direct verification; this is the classical multiplier family, cf. [16, 4]). Define

$$W(n) = \min\{M \equiv 3\ (4) :\ n \equiv -uv^{-1} \pmod M \text{ for some } uvw = A_M\}, \tag{2}$$

with $W(n) = +\infty$ if no such $M$ exists. Thus $W(p) \le T$ supplies an explicit representation with a modulus at most $T$.

*The slice parameter.* A Type I solution with parameters $(a,b,c,k)$ satisfies $p(a+b) = k(4abc-1)$ and gives $4/p = 1/(ack) + 1/(bck) + 1/(pabc)$. For an admissible pair $(c,k)$ (see §14) let $M_{c,k}(p)$ be the number of divisors of $p^2+4ck^2$ lying in the class $-p \pmod{4ck}$; the slice is positive iff $M_{c,k}(p) > 0$ (see §14). Put $ck_{\min}(p) = \min\{ck :\ M_{c,k}(p) > 0,\ \mathrm{sf}(c) \notin \{1,2,3,6\}\}$ (the cores $1,2,3,6$ are excluded because they are deterministic on hard primes).

Both parameters are expected to be small for almost all primes. For $W$, tails of the form $\#\{p \le N : W(p) > T\} \ll N \exp\{-c(\log T)^2 \log\log T\}$ (in the range $\log N \gg (\log T)^3 \log\log T$) are obtained in companion work whose assembly is still under internal review; we do not use them. In the other direction, the class of one (every prime $p \equiv 1 \pmod{\mathrm{lcm}\{M \le T\}}$ has $W(p) > T$) together with Linnik's theorem shows $W(p) \gg \log p$ infinitely often. With the smaller modulus $\mathrm{lcm}(24,\ M \le T,\ M \equiv 3\ (4))$ and Chang's least-prime bound for smooth moduli [2], the constant becomes $5/8$ (§13). The same is true for $ck_{\min}$ via genus characters. Whether $W(p)/\log p$ is unbounded was open.

**Theorem 1.1** (Proved modulo Theorem 4.1; effective for each fixed $k$)**.** *For every integer $k \ge 3$ there is a constant $C_k$ such that for infinitely many hard primes $p$*

$$W(p) \ \ge\ (\log p)^k \exp\Big(-C_k \frac{\log\log p}{\log\log\log p}\Big).$$

*More precisely, for every $k \ge 3$ and every sufficiently large $T$ there is a prime $p \equiv 1 \pmod{840}$ with $W(p) > T$ and $\log p \le T^{1/k} \exp(O_k(\log T/\log\log T))$. In particular the least hard prime $p$ with $W(p) > T$ is at most $\exp(T^{o(1)})$.*

In particular the statement "$W(p) \le (\log p)^A$ for all large $p$" is false for every $A$. The cited input (Theorem 4.1) is Thorner and Zaman's prime

number theorem in the Linnik range [15], including the McCurley zero-free region statement quoted there. No hypothesis on Siegel zeros is needed: the exceptional zero is absorbed by a uniqueness lemma across the family of moduli (Lemma 4.2) and a two-case analysis (Theorem 5.1).

The proof has two layers. The analytic layer (Theorem 5.1) turns any congruence minorant of $\mathbf{1}[W(n) > T]$ with controlled modulus, $\ell^1$-mass and twisted means into a prime. The combinatorial layer produces such minorants. The simplest one is *prime-local*: the class of one modulo all primes up to $T^{1/2+o(1)}$ leaves a single forbidden set at each larger prime, and a Brun minorant applies; this gives exponent 2 (Theorem 6.1), which is the limit of every prime-local design (Proposition 13.3). To go further we quarantine only the primes up to $y = T^{1/k+o(1)}$ (plus $T^{o(1)}$ further primes, chosen by an iterated quarantine so that every free prime carries little mass); then moduli with up to $k-1$ free prime factors produce events on up to $k-1$ primes. These cluster at "hub" residues on which many events fire together, so the usual Bonferroni truncation by the number of events fails (Proposition 13.6). We truncate instead by the number of *primes* in the support (§7). For events on at most two primes the error is controlled by a pseudoforest bound after quarantining high-degree hub residues (§8); this already gives exponent 3 (the case $k = 3$). For larger supports a hypergraph moment bound (§10) needs *codegree* conditions at the scale $1/L$ of the truncation level $L$, and some vertex pairs (the classes of small-height rationals such as $-4$, $-4d^2$, $-2/3$) genuinely violate them. The remedy (§11) is to sieve in levels: events on $r$ primes form level $r$, the graph level 2 is outermost, and each level is composed inside the lower ones coefficient by coefficient. A conditional local lemma shows that the error of level $r$ is relative to the probability of avoiding the lower levels, so its truncation, and hence its codegree threshold, depends only on its own mass; heavy vertex sets are moved down to the level of their size by Markov's inequality, and level 2 needs no codegree condition. No arithmetic bound on codegrees is used. The prime side thus matches the Haar (profinite) side, where an iterated quarantine gives $\log(1/\delta^*(T)) \le T^{o(1)}$ (Theorem 15.9).

For the slice parameter the picture is different: the only congruence mechanism is the genus character, and it certifies exactly the least quadratic non-residue $n_p$.

**Theorem 1.2** (Proved; (b) modulo [8, Prop. 5.1])**.**

(a) (Proved.) $ck_{\min}(p) \ge n_p$ *for every prime* $p \equiv 1 \pmod{24}$*, and every fixed-period congruence minorant for* $\{ck_{\min}(p) > T\}$ *is supported, among large primes, on primes with* $n_p > T$ (Corollary 14.5).

(b) (Proved modulo [8, Prop. 5.1].) $ck_{\min}(p) \gg \log p \cdot \log\log\log p$ *for infinitely many hard primes.*

Thus a congruence proof of $ck_{\min}(p) > (\log p)^{1+\delta}$ infinitely often would improve the best known unconditional (Graham–Ringrose) and GRH-conditional

(Montgomery) $\Omega$-results for $n_p$, and would exceed the random-model prediction $\max_{p\le x} n_p \asymp \log x \log\log x$.

Finally, on the Haar (profinite) side, let $\delta^*(T)$ be the Haar measure of the set of $n \in \widehat{\mathbb{Z}}^\times$ with $n \equiv 1 \pmod{24}$ and $W(n) > T$, normalised within the class $1 \pmod{24}$. The class of one gives $\log(1/\delta^*(T)) \le (2/3 + o(1))T$, and a crude prime-local bound gives $T^{1/2+o(1)}$.

**Theorem 1.3** (Proved; the polylogarithmic form modulo [4, Prop. 1.4])**.** $\log(1/\delta^*(T)) \le T^{o(1)}$*; more precisely* $\log(1/\delta^*(T)) \ll (\log T)^3(S^* + 1)$*, where* $S^* \le \exp(O(\log T/\log\log T))$ *is the uniform atom mass of §12, and* $\log(1/\delta^*(T)) \ll (\log T)^7/\log\log T$ *modulo* [4, Prop. 1.4].

This is Theorem 15.9, proved in §15 by an iterated quarantine and the local lemma; an elementary warm-up gives $T^{1/3+o(1)}$ (Theorem 15.3).

We do not pursue heuristics about the true size of $\delta^*(T)$ here.

**Status conventions.** Every theorem header carries a label. *Proved* means proved in full here. *Proved modulo X* means proved here assuming the cited statement X, whose statement we checked in the source but whose proof we did not check. *Proved modulo the proof of X* means that we use a bound established inside the proof of X, as stated there. *Cited* marks external statements. *Classical* marks a standard result for which we include a short proof from a standard reference. *Proved implication* means that the implication from an explicitly stated, open hypothesis is proved in full. *Assessment* marks heuristic discussion that is not a theorem.

## 2. The atoms of the multiplier problem

For $M \equiv 3 \pmod 4$ put $\mathcal{R}(M) = \{-4D \bmod M : \ D \mid A_M^2\}$.

**Lemma 2.1** (Proved)**.** $\{-uv^{-1} \bmod M : \ uvw = A_M\} = \mathcal{R}(M)$*. Consequently, for every integer* $n$*,*

$$W(n) > T \iff n \bmod M \notin \mathcal{R}(M) \text{ for every } M \le T,\ M \equiv 3\ (4).$$

*Proof.* Since $4A_M \equiv 1 \pmod M$ and $\gcd(A_M, M) = 1$, from $uvw = A_M$ we get $v^{-1} \equiv 4uw$, hence $-uv^{-1} \equiv -4u^2w$ and $D = u^2w$ divides $A_M^2$. Conversely let $D \mid A_M^2$. For each prime $q$ with $q^a \| A_M$ and $q^d \| D$ $(d \le 2a)$ put $u_q = \max(0, d-a)$, $w_q = d - 2u_q$, $v_q = a - u_q - w_q$; these are non-negative, so $u = \prod q^{u_q}$, $v = \prod q^{v_q}$, $w = \prod q^{w_q}$ satisfy $uvw = A_M$ and $u^2w = D$. □

**Lemma 2.2** (Class of one; proved)**.** $1 \notin \mathcal{R}(M)$ *for every* $M \equiv 3 \pmod 4$.

*Proof.* If $-4D \equiv 1$ then $D \equiv -A_M \pmod M$. The map $D \mapsto A_M^2/D$ preserves this congruence, so we may assume $D \le A_M$; then $0 < D + A_M \le 2A_M < M$ while $M \mid D + A_M$. □

## 3. The prime-local residual system

Fix $T$ large and put $\mathcal{L} = \log T$,

$$y = \sqrt{T}\,\exp(3\mathcal{L}/\log\mathcal{L}),$$
$$Q = \operatorname{lcm}\big(24,\ \ell^{e_\ell} :\ \ell \le y \text{ prime},\ \ell^{e_\ell} \le T \text{ maximal}\big),$$

and let $\mathcal{U}$ be the set of primes in $(y, T]$. Then $\log Q \le 2y$ for large $T$ (only $\ell \le \sqrt{T}$ have $e_\ell \ge 2$), and $840 \mid Q$. For $\ell \in \mathcal{U}$ let

$$F_\ell = \{-4D \bmod \ell :\ m \le T/\ell,\ m\ell \equiv 3\ (4),\ D \mid A_{m\ell}^2,\ m \mid 4D+1\},$$
$$g(\ell) = \frac{|F_\ell|}{\ell - 1}, \qquad S = \sum_{\ell\in\mathcal{U}} g(\ell).$$

All elements of $F_\ell$ are units.

**Lemma 3.1** (Prime-local reduction; proved)**.** *If $n \equiv 1 \pmod{Q}$ and $n \bmod \ell \notin F_\ell$ for all $\ell \in \mathcal{U}$, then $W(n) > T$.*

*Proof.* Let $M \le T$, $M \equiv 3 \pmod 4$. If every prime factor of $M$ is at most $y$, then $M \mid Q$, so $n \equiv 1 \pmod M$ and Lemma 2.2 applies. Otherwise let $\ell > y$ divide $M$; since $\ell^2 > T$ we have $\ell \| M$, and $m = M/\ell < T/y \le y$, so $m \mid Q$ and $n \equiv 1 \pmod m$. If $n \equiv -4D \pmod M$ with $D \mid A_M^2$, then $m \mid 4D+1$ and $n \bmod \ell \in F_\ell$. Now use Lemma 2.1. □

**Lemma 3.2** (Local sizes; proved)**.** *$g(\ell) \le 2T\tau^*(T)^2/y^2$, where $\tau^*(T) = \max_{n\le T}\tau(n)$. In particular $\max_\ell g(\ell) \le 1/16$ for large $T$.*

*Proof.* For each $m \le T/\ell$ there are at most $\tau(A^2) \le \tau(A)^2$ choices of $D$, and $1/(\ell-1) \le 2/\ell$. Since $\log\tau^*(T) \le (\log 2 + o(1))\mathcal{L}/\log\mathcal{L}$ [7, Thm. 317], the bound is $\le 2\exp((2\log 2 - 6 + o(1))\mathcal{L}/\log\mathcal{L})$. □

The next lemma is the key arithmetic input: the congruence $m \mid 4D+1$ saves a factor $m$.

**Lemma 3.3** (Mass bound; proved)**.** *With $X = (T+1)/4$,*

$$S \le \tfrac{4}{3}(3 + \log X) \sum_{\substack{sr^2 \le X \\ s\ \text{squarefree}}} \frac{\tau(4sr^2+1)}{sr}$$
$$\le 2(3+\mathcal{L})(1+\mathcal{L})^2\tau^*(T+2) = \exp\Big(O\Big(\frac{\log T}{\log\log T}\Big)\Big).$$

*Proof.* Clearly $S \le \sum 1/(\ell-1)$ over triples $(m,\ell,D)$ as in the definition of $F_\ell$; put $A = A_{m\ell}$. Since $4A \equiv 1$ and $4D \equiv -1 \pmod m$, we have $4A^2D^{-1} \equiv -1 \pmod m$, so $D \mapsto A^2/D$ preserves $m \mid 4D+1$ and exchanges $D > A$ with $D < A$; hence it suffices to count $D \le A$ and double. Write $D = sr^2$ with $s$ squarefree. Then $D \mid A^2$ iff $sr \mid A$; write $A = srk$, so that $D \le A$ iff $k \ge r$. The map $(m,\ell,D) \mapsto (m,s,r,k)$ is injective, and $1/(\ell-1) \le 2/\ell = 2m/(4A-1) \le 2m/(3A)$.

From $m \mid 4sr^2+1$ and $m \mid m\ell = 4srk-1$ we get $m \mid 4sr(r+k)$, and $\gcd(m, 4sr) = 1$, so

$$m \mid r+k.$$

Since $k \geq r \geq 1$, the least admissible $k$ is at least $\max(r, m-r) \geq m/2$, so $\sum_{k \equiv -r\,(m),\ k \leq X} 1/k \leq (3+\log X)/m$. Summing over $m \mid 4sr^2+1$ gives the first bound; the second follows from $\sum_{s \leq X} 1/s \cdot \sum_{r \leq \sqrt{X}} 1/r \leq (1+\mathcal{L})^2$ and $4sr^2+1 \leq T+2$. □

**Remark 3.4.** Without the congruence $m \mid r+k$ the same count gives $S \leq T^{1/2-\eta+o(1)}$ for $y = T^{1/2+\eta}$, and the transfer below then only yields $W(p) \gg \log p$. Numerically $S \approx 6.5,\ 13.6,\ 23.9,\ 38.5$ for $T = 10^3, \ldots, 10^6$ (with $y = \sqrt{T}$), i.e. $S$ is polylogarithmic in practice. These values illustrate $S$ only: the parameter $y = \sqrt{T}$ does *not* instantiate Theorem 6.1. At $y = \sqrt{T}$ one has $\max g \approx 0.08$–$0.33 > 1/16$, and the twist condition of §6 can fail (numerically it does at $T = 10^4$). The bound $g \leq 1/16$ of Lemma 3.2, and hence Theorem 6.1, is asymptotic: with the $y$ above, $y < T$ (so that $\mathcal{U} \neq \emptyset$) only when $\log T > e^6 \approx 403$.

## 4. The analytic input

**Theorem 4.1** (Thorner–Zaman [15, Cor. 1.4 and Rem. 1.5]; cited)**.** *There are absolute effectively computable constants $c_4 > 0$, $C_0$ such that for coprime $a, q$ with $q \geq 2$ and $x \geq q^{12}$,*

$$\sum_{\substack{p \leq x \\ p \equiv a\,(q)}} \log p = \frac{\lambda x}{\varphi(q)}\,(1+\varepsilon),$$

$$|\varepsilon| \leq C_0\Big(e^{-c_4 \log x/\log q} + e^{-c_4 (\log x)^{3/5}/(\log\log x)^{1/5}}\Big),$$

*where $\lambda = 1 - \chi_1(a) x^{\beta_1 - 1}/\beta_1$ if the exceptional zero $\beta_1$ for the modulus $q$ exists and $\lambda = 1$ otherwise. Here, following McCurley [9] as quoted in* [15, p. 1]*, $\prod_{\chi \bmod q} L(s,\chi)$ has no zero in $\Re s \geq 1 - 1/(13\log(q(|\Im s|+3)))$ apart from at most one real zero $\beta_1$, which is simple and belongs to a unique real character $\chi_1$; also $0 < \lambda < 2$.*

We used the arXiv version; we did not compare the numbering with the published version. We also use the quoted McCurley statement for moduli other than those of Theorem 4.1; alternatively the classical Landau–Page theorem [3, Ch. 14] with a constant $c \leq 1/13$ in place of $1/13$ works verbatim.

**Lemma 4.2** (One severe character per family; proved modulo the McCurley statement)**.** *Let $\mathcal{Q}$ be a finite set of moduli, all $\leq Z$. Call $(\chi^*, \beta)$* severe *if $\chi^*$ is a real primitive character whose conductor $q^*$ divides some $q \in \mathcal{Q}$, $L(\beta, \chi^*) = 0$ and $\beta \geq 1 - 1/(13\log(3Z^2))$. Then there is at most one severe pair. If $(\chi^*, \beta^*)$ is severe and $q^* \mid q \in \mathcal{Q}$, then $\beta_1 = \beta^*$ and $\chi_1$ is induced by $\chi^*$ for the modulus $q$. If $q \in \mathcal{Q}$ and no severe pair has $q^* \mid q$, then any exceptional $\beta_1$ of $q$ satisfies $x^{\beta_1 - 1}/\beta_1 \leq 2\exp(-\log x/(13\log 3Z^2))$.*

*Proof.* If $(\chi_a,\beta_a),(\chi_b,\beta_b)$ are severe with conductors dividing $q_a,q_b\in\mathcal{Q}$, put $q'=\operatorname{lcm}(q_a,q_b)\le Z^2$. Both characters induce characters mod $q'$ with the same zeros in $\Re s>0$, and both $\beta$ lie in McCurley's region for $q'$; uniqueness there gives $\beta_a=\beta_b$ and $\chi_a=\chi_b$. The second claim holds because $\beta^*$ lies in McCurley's region for $q\le Z$. For the third, an exceptional zero of $q$ in $[1-1/(13\log 3Z^2),1)$ would be severe; also $\beta_1\ge 1-1/(13\log 6)>1/2$. □

## 5. A transfer theorem

**Theorem 5.1** (Proved modulo Theorem 4.1)**.** *Let $T\ge 1$, $Q\ge 2$, and let $B(n)=\sum_{i\in I}c_i\,\mathbf{1}[n\equiv b_i\ (d_i)]$ be a finite real combination with $\gcd(d_i,Q)=\gcd(b_i,d_i)=1$ and $B(n)\le\mathbf{1}[W(n)>T]$ for all $n\equiv 1\pmod Q$. Put $Z=Q\max d_i$, $\mu=\sum c_i/\varphi(d_i)$, $M_1=\sum|c_i|/\varphi(d_i)$, and assume $\mu>0$ and the* twist condition*: for every real primitive character $\psi$ of conductor $f>1$ with $\gcd(f,Q)=1$ and $f\mid d_i$ for some $i$,*

$$|\mu_\psi|\le\mu/4,\qquad \mu_\psi=\sum_{i:\,f\mid d_i}c_i\psi(b_i)/\varphi(d_i).$$

*Let $K=1+\log(M_1/\mu)$. There is an absolute $C_1$ such that if $\log x\ge C_1K\max(\log Z,K)$, then some prime $p\le x$, $p\equiv 1\pmod Q$, has $W(p)>T$.*

*Proof.* Let $q_i=Qd_i$ and let $a_i$ be the class $\equiv 1\ (Q)$, $\equiv b_i\ (d_i)$; it is a unit class. Then $\sum_{p\le x,\,p\equiv 1(Q)}\log p\,B(p)=\sum_i c_i\theta(x;q_i,a_i)$. Since $M_1\ge\mu$ we have $K\ge 1$, so taking $C_1\ge 12$ gives $\log x\ge 12\log Z$, i.e. $x\ge Z^{12}\ge q_i^{12}$, and Theorem 4.1 applies to every term. Put $E_1=e^{-c_4\log x/\log Z}+e^{-c_4(\log x)^{3/5}/(\log\log x)^{1/5}}$ and $E_2=e^{-\log x/(13\log 3Z^2)}$. Since $\gcd(Q,d_i)=1$ we have $\varphi(q_i)=\varphi(Q)\varphi(d_i)$, and $q_i\le Z$, so

$$\sum_i c_i\theta(x;q_i,a_i)=\frac{x}{\varphi(Q)}\sum_i\frac{c_i\lambda_i(1+\varepsilon_i)}{\varphi(d_i)},\qquad |\varepsilon_i|\le C_0E_1,$$

with $\lambda_i$ the factor of Theorem 4.1 for $(q_i,a_i)$. Apply Lemma 4.2 to $\mathcal{Q}=\{q_i\}$; a severe $\beta$ satisfies $\beta\ge 1-1/(13\log 6)>1/2$. Write the conductor of a severe $\chi^*$ as $q^*=q^*_Qq^{*\prime}$ with $q^*_Q$ supported on the primes of $Q$ and $\gcd(q^{*\prime},Q)=1$, and $\chi^*=\chi^*_Q\chi^{*\prime}$.

*Case A: a severe pair exists with $q^*\mid Q$.* Then $q^*\mid q_i$ for every $i$. By the second claim of Lemma 4.2, every $\lambda_i$ is formed with the same pair $(\chi^*,\beta^*)$, so $\lambda_i=1-\chi^*(a_i)x^{\beta^*-1}/\beta^*=1-x^{\beta^*-1}/\beta^*=:\lambda_*$, because $a_i\equiv 1\pmod{q^*}$. Moreover $\lambda_*>0$: since $\log x\ge 12\log Z>2$ and $1/2<\beta^*<1$, $\log\beta^*\ge 1-1/\beta^*>2(\beta^*-1)>(\beta^*-1)\log x$, i.e. $x^{\beta^*-1}/\beta^*<1$. So $\sum c_i\theta\ge(x\lambda_*/\varphi(Q))(\mu-C_0E_1M_1)$.

*Case B: otherwise.* If a severe $\chi^*$ exists, then $q^*$ divides some $Qd_j$ with $\gcd(d_j,Q)=1$, so $q^*_Q\mid Q$. Hence $q^*\mid q_i$ iff $q^{*\prime}\mid d_i$, and in Case B we have $q^{*\prime}>1$. For such $i$, $\chi_1=\chi^*$ on units mod $q_i$ (Lemma 4.2), and, as $a_i\equiv 1\pmod{q^*_Q}$ and $a_i\equiv b_i\pmod{q^{*\prime}}$, $\chi^*(a_i)=\chi^*_Q(1)\chi^{*\prime}(b_i)=\psi(b_i)$ with $\psi=\chi^{*\prime}$. So $\lambda_i=1-\psi(b_i)x^{\beta^*-1}/\beta^*$. This $\psi$ is real primitive (the coprime-conductor factors of a real primitive character are real primitive), of conductor $q^{*\prime}>1$

coprime to $Q$ and dividing some $d_i$, so the twist condition applies to it. For all other $i$ (and for all $i$ if there is no severe pair), $|\lambda_i - 1| \le x^{\beta_1 - 1}/\beta_1 \le 2E_2$ by the third claim of Lemma 4.2. Since $\lambda_i < 2$ and $x^{\beta^* - 1}/\beta^* \le 1$, splitting $\lambda_i = 1 + (\lambda_i - 1)$ gives

$$\begin{aligned}\sum c_i\theta &\ge \frac{x}{\varphi(Q)}\big(\mu - 2|\mu_\psi| - M_1(2C_0E_1 + 2E_2)\big)\\ &\ge \frac{x}{\varphi(Q)}\big(\mu/2 - M_1(2C_0E_1 + 2E_2)\big),\end{aligned}$$

using $|\mu_\psi| \le \mu/4$. Both brackets are positive once $E_1, E_2 \le \mu/(8(C_0 + 1)M_1) = e^{1-K}/(8(C_0+1))$. This holds if $C_1$ is large: $\log x/\log Z \ge C_1K$, $(\log x)^{3/5}/(\log\log x)^{1/5} \ge (\log x)^{1/2} \ge C_1^{1/2}K$ and $\log x/(13\log 3Z^2) \ge C_1K/52$ (as $Z \ge 2$), so each of the three exponentials is at most $e^{-cC_1^{1/2}K}$ with $c > 0$ absolute. Hence $\sum_p \log p\, \mathbf{1}[W(p) > T] \ge \sum_p \log p\, B(p) > 0$. □

No lower bound for $1 - \beta^*$ is used: in Case A the exceptional factor $\lambda_*$ multiplies main term and error alike. A prime number theorem valid only for $q \le \exp(c\sqrt{\log x})$ would force $\log x \gg (\log Z)^2 \ge T$; the Linnik-range uniformity is essential.

## 6. Warm-up: exponent two by a prime-local design

**Theorem 6.1** (Proved modulo Theorem 4.1; effective)**.** *For infinitely many hard primes $p$, $W(p) \ge (\log p)^2 \exp(-C\log\log p/\log\log\log p)$. More precisely, for every large $T$ there is a prime $p \equiv 1 \pmod Q$ ($Q$ as in §3) with $W(p) > T$ and $\log p \le T^{1/2}\exp(O(\log T/\log\log T))$.*

Theorem 1.1 supersedes this; we include it because its minorant is a plain Brun truncation and it isolates the arithmetic input (Lemma 3.3) that the main proof reuses.

*Proof.* Let $J$ be the least even integer $\ge 22S + 10$ and

$$B(n) = \sum_{P \subseteq \mathcal{U},\ |P| \le J-1} (-1)^{|P|} \prod_{\ell \in P} \mathbf{1}[n \bmod \ell \in F_\ell] = \sum_{j=0}^{J-1} (-1)^j \binom{N(n)}{j},$$
$$N(n) = \#\{\ell \in \mathcal{U} :\ n \bmod \ell \in F_\ell\}.$$

*Pointwise bound.* For $N \ge 1$, $\sum_{j \le J-1}(-1)^j\binom{N}{j} = (-1)^{J-1}\binom{N-1}{J-1} \le 0$; hence $B \le \mathbf{1}[N = 0] \le \mathbf{1}[W > T]$ on $n \equiv 1\ (Q)$ by Lemma 3.1. Expanding $\mathbf{1}[n \bmod \ell \in F_\ell] = \sum_{a \in F_\ell} \mathbf{1}[n \equiv a\ (\ell)]$ puts $B$ in the form of Theorem 5.1, with squarefree moduli $d_P = \prod_{\ell \in P}\ell \le T^J$ coprime to $Q$.

*Mean and mass.* With $e_j$ the elementary symmetric functions of the $g(\ell)$, $\mu = \sum_{j<J}(-1)^j e_j$ and $M_1 = \sum_{j<J} e_j \le e^S$. Let $N$ be a sum of independent Bernoulli$(g(\ell))$ variables; the identity above and $\binom{N-1}{J-1} \le \binom{N}{J}$ give $|\mu - V| \le e_J \le (eS/J)^J \le e^{-2J}$, where $V = \prod(1 - g(\ell)) \ge e^{-2S}$ (Lemma 3.2). So $\mu \ge 0.99V$ and $\log(M_1/\mu) \le 3S + 1$.

*Twist.* If $\psi$ is real primitive of conductor $f>1$, coprime to $Q$, dividing some $d_P$, then $f=\prod_{\ell\in P_f}\ell$ with $r=|P_f|\ge 1$ and $\psi=(\cdot/f)$. With $h(\ell)=\sum_{a\in F_\ell}(a/\ell)/(\ell-1)$, $|h|\le g$,

$$\mu_\psi=(-1)^r\prod_{\ell\in P_f}h(\ell)\sum_{P'\subseteq\mathcal{U}\setminus P_f,\ |P'|\le J-1-r}(-1)^{|P'|}\prod_{P'}g,$$

which vanishes if $r\ge J$. The inner sum is $V'+\varepsilon'$ with $V'=V/\prod_{P_f}(1-g)\le(16/15)^rV$ and $|\varepsilon'|\le e_{J-r}\le S^{J-r}/(J-r)!$, so, with $g\le 1/16$, $|\mu_\psi|\le 15^{-r}V+16^{-r}S^{J-r}/(J-r)!$. The first term is $\le V/15$. If $r\le J/2$, the second is $\le(eS/(J-r))^{J-r}\le(2eS/J)^{J/2}\le(e/11)^{J/2}\le e^{-0.69J}\le e^{-15S-6}$, as $\frac12\log(11/e)>0.69$ and $J\ge 22S+10$. If $r>J/2$, it is $\le 16^{-J/2}e^S\le e^{-1.38(22S+10)+S}\le e^{-15S-6}$. Since $e^{-15S}\le e^{-2S}\le V$, we get $|\mu_\psi|\le(1/15+e^{-6})V\le 0.07V\le\mu/4$.

*Moduli.* $\log Z\le\log Q+J\mathcal{L}\le 3y$ because $S=T^{o(1)}$ (Lemma 3.3). Theorem 5.1 gives a prime $p\equiv 1\pmod Q$ with $W(p)>T$ and $\log p\le C_1(3S+2)\cdot 3y\le\sqrt{T}\exp((3+\log 2+o(1))\mathcal{L}/\log\mathcal{L})$. Since $840\mid Q$ the prime is hard, and $p>Q>T$ gives infinitely many distinct primes. Inverting: if $\mathcal{L}\ge 3\log\log p$ then already $W(p)>T\ge(\log p)^3$; otherwise $\mathcal{L}/\log\mathcal{L}\ll\log\log p/\log\log\log p$ and $\mathcal{L}\ge 2\log\log p-O(\log\log p/\log\log\log p)$. All inputs are effective. □

## 7. Support-truncated inclusion–exclusion

*Setting.* Let $\mathcal{P}$ be a finite set of primes and $X=(X_\ell)_{\ell\in\mathcal{P}}$ independent random variables. Let $\mathcal{E}$ be a finite family of events; each $E\in\mathcal{E}$ has a nonempty support $\operatorname{supp}E\subseteq\mathcal{P}$ and depends only on $(X_\ell)_{\ell\in\operatorname{supp}E}$. For an outcome $x$ put

$$A(x)=\{E\in\mathcal{E}:\ E\text{ occurs}\},\quad V(x)=\bigcup_{E\in A(x)}\operatorname{supp}E,\quad N(x)=|V(x)|,$$

$$a_\ell(x)=\#\{E\in A(x):\ \ell\in\operatorname{supp}E\}.$$

For $L\ge 0$ and $F\subseteq\mathcal{E}$ with $\operatorname{supp}F=\bigcup_{E\in F}\operatorname{supp}E$, define

$$B_L(x)=\sum_{F\subseteq A(x),\ |\operatorname{supp}F|\le L}(-1)^{|F|},$$

$$G_{L+1}(x)=e_{L+1}(a(x))=\sum_{P\subseteq\mathcal{P},\ |P|=L+1}\prod_{\ell\in P}a_\ell(x).$$

The truncation is by the number of *primes* involved, not by the number of events; this is what defeats the hub configurations of Proposition 13.6.

**Lemma 7.1** (Proved)**.** *Call $W\subseteq\mathcal{P}$ independent at $x$ if no $E\in A(x)$ has $\operatorname{supp}E\subseteq W$. If $A(x)=\emptyset$ then $B_L(x)=1$. Otherwise*

$$B_L(x)=\sum_{\substack{W\subsetneq V(x)\\ \text{independent at } x\\ |W|\le L}}(-1)^{L-|W|}\binom{N-|W|-1}{L-|W|},$$

*and in particular* $B_L(x) = 0$ *whenever* $1 \le N(x) \le L$.

*Proof.* Fix $x$, write $A, V, N$. For $U \subseteq V$ the alternating sum over all $F \subseteq \{E \in A : \operatorname{supp} E \subseteq U\}$ is $I(U) = \mathbf{1}[\text{no } E \in A \text{ has } \operatorname{supp} E \subseteq U]$, so by Möbius inversion on the Boolean lattice $\sum_{F \subseteq A,\ \operatorname{supp} F = U} (-1)^{|F|} = \sum_{W \subseteq U} (-1)^{|U \setminus W|} I(W)$. Summing over $U \subseteq V$ with $|U| \le L$,

$$B_L = \sum_{W \subseteq V,\ I(W)=1,\ |W| \le L} \ \sum_{j=0}^{L-|W|} (-1)^j \binom{N - |W|}{j}.$$

The inner sum is $(-1)^{L-|W|}\binom{N-|W|-1}{L-|W|}$ if $|W| < N$ and 1 if $W = V$; and $I(V) = 1$ iff $A = \emptyset$. If $N \le L$ every binomial $\binom{N-|W|-1}{L-|W|}$ vanishes. □

**Lemma 7.2** (Proved)**.** *For every* $x$ *and* $L \ge 0$,

$$B_L^*(x) := B_L(x) - 4^{L+1} G_{L+1}(x) \le \mathbf{1}[A(x) = \emptyset],$$
$$|B_L(x) - \mathbf{1}[A(x) = \emptyset]| \le 4^{L+1} G_{L+1}(x).$$

*Proof.* If $A = \emptyset$ then $a \equiv 0$ and $B_L = 1$. Let $A \neq \emptyset$. If $N \le L$ then $B_L = 0$. If $N \ge L + 1$, Lemma 7.1 and Vandermonde give $|B_L| \le \sum_{w \le L} \binom{N}{w}\binom{N}{L-w} = \binom{2N}{L}$. For $f(N) = \binom{2N}{L}/\binom{N}{L+1}$ one computes

$$\frac{f(N+1)}{f(N)} = \frac{2(2N+1)(N-L)}{(2N+2-L)(2N+1-L)},$$

and with $t = N - L \ge 1$ the denominator exceeds the numerator by $L^2 + 3L + 4t + 2$; so $\binom{2N}{L} \le f(L+1)\binom{N}{L+1} \le 4^{L+1}\binom{N}{L+1}$. Finally $\binom{N}{L+1} = e_{L+1}(\mathbf{1}[\ell \in V]) \le e_{L+1}(a)$. □

For a finite combination $\Phi = \sum_i c_i \mathbf{1}_{\mathcal{C}_i}$ of indicators of cells $\mathcal{C}_i$ (a cell being a product of single values of finitely many coordinates) we write $M_1(\Phi) = \sum_i |c_i| P(\mathcal{C}_i)$, after merging equal cells; for congruence classes this is the $M_1$ of Theorem 5.1.

**Lemma 7.3** (Proved)**.** *Let* $z \ge 16$ *and* $\Lambda_z = \log E \prod_{\ell \in \mathcal{P}} (1 + z a_\ell)$. *Then* $E\, G_{L+1} \le z^{-L-1} e^{\Lambda_z}$ *and* $E|B_L^* - \mathbf{1}[A = \emptyset]| \le 2 \cdot 4^{L+1} z^{-L-1} e^{\Lambda_z}$; *for* $z = 16$ *this is* $2 \cdot 4^{-L-1} e^{\Lambda_{16}}$. *If every event is a union of cells, then* $B_L$ *is a combination of cells on supports of size* $\le L$ *with* $M_1(B_L) \le E \prod_\ell (1 + 2a_\ell)$, *and* $G_{L+1}$ *is a nonnegative combination of cells on supports of size* $\le k(L+1)$, $k = \max |\operatorname{supp} E|$, *with* $M_1(G_{L+1}) = E\, G_{L+1}$. *If moreover no event contains a cell with a non-unit coordinate, then only unit cells have nonzero coefficient.*

*Proof.* Since all $a_\ell \ge 0$, we have $e_{L+1}(a) \le z^{-L-1} \prod_\ell (1 + z a_\ell)$ termwise. By Lemma 7.2,

$$E|B_L^* - \mathbf{1}[A = \emptyset]| \le E|B_L - \mathbf{1}[A = \emptyset]| + 4^{L+1} E\, G_{L+1} \le 2 \cdot 4^{L+1} E\, G_{L+1}.$$

Write $B_L = \sum_{F \subseteq \mathcal{E},\, |\operatorname{supp} F| \le L} (-1)^{|F|} \prod_{E \in F} \mathbf{1}_E$ and group by $U = \operatorname{supp} F$. On a cell $c$ of the coordinates in $U$, let $A_U(c)$ be the set of events with support

in $U$ that contain $c$, and $I_c(W) = \mathbf{1}[\text{no } E \in A_U(c) \text{ has } \operatorname{supp} E \subseteq W]$. The coefficient of $c$ is

$$\kappa(U,c) = \sum_{F \subseteq A_U(c),\ \operatorname{supp} F = U} (-1)^{|F|} = \sum_{W \subseteq U} (-1)^{|U \setminus W|} I_c(W)$$

by the Möbius inversion of Lemma 7.1; so $|\kappa(U,c)| \le 2^{|U|}$. If some $\ell \in U$ lies in the support of no event of $A_U(c)$, then $I_c(W) = I_c(W \cup \{\ell\})$ and the terms cancel in pairs, so $\kappa(U,c) = 0$; in particular $\kappa(U,c) = 0$ unless $c$ is a unit cell, under the last hypothesis. For the remaining cells, every $\ell \in U$ is covered by an event occurring on $c$, so $\mathbf{1}_c \le \prod_{\ell \in U} a_\ell$ on $c$. Hence

$$\begin{aligned} M_1(B_L) &\le \sum_U 2^{|U|} P(\text{every } \ell \in U \text{ covered}) \\ &\le \sum_U 2^{|U|} E \prod_{\ell \in U} a_\ell = E \prod_\ell (1 + 2a_\ell). \end{aligned}$$

Finally $G_{L+1} = \sum_{|P|=L+1} \prod_{\ell \in P} \sum_{E \ni \ell} \mathbf{1}_E$ expands into products of event indicators with coefficient $+1$, so its mass equals its mean. □

## 8. A pseudoforest bound for the exponential moment

*Graph-type systems.* Each $X_\ell$ takes values in a finite set $\Omega_\ell$. The events are of two kinds. *Singles*: one set $S_\ell \subseteq \Omega_\ell$ per $\ell$, with $g_\ell = P(X_\ell \in S_\ell)$ and $S_1 = \sum_\ell g_\ell$. *Edges*: a *vertex* is a pair $v = (\ell, V_v)$ with $V_v \subseteq \Omega_\ell$, distinct vertices at the same $\ell$ being disjoint, and $p(v) = P(X_\ell \in V_v)$; an edge is an unordered pair $\{u, w\}$ of vertices at distinct primes, occurring iff $X_{\ell_u} \in V_u$ and $X_{\ell_w} \in V_w$. The edge graph is simple: edges are distinct pairs of vertices (for congruences, distinct classes modulo $\ell\ell'$), repeated atoms being merged. Put

$$S_2 = \sum_{\text{edges}} p(u)p(w), \quad \deg(v) = \sum_{w:\{v,w\} \text{ edge}} p(w), \quad w_\ell = \sum_{\text{edges at } \ell} p(u)p(w).$$

Then $a_\ell = s_\ell + d_\ell$ with $s_\ell = \mathbf{1}[X_\ell \in S_\ell]$ and $d_\ell$ the number of occurring edges at $\ell$.

**Lemma 8.1** (Proved)**.** *Let $z \ge 1$ and suppose $\deg(v) \le \delta \le e^{-3z-2}$ for every vertex. Then $\Lambda_z \le zS_1 + 16e^{6z+2}S_2$.*

*Proof.* Use $1 + z(s+d) \le (1+zs)(1+zd)$ and expand $\prod_\ell (1 + zd_\ell) = \sum_{(U,f)} z^{|U|} \mathbf{1}[f(\ell) \text{ occurs for all } \ell \in U]$, where $f$ assigns to each $\ell \in U$ an edge at a vertex of $\ell$. Let $F = f(U)$. If $F$ occurs, its vertices lie at distinct primes, and each component $K$ of $F$ has $|E(K)| \le |V(K)|$ (each edge of $K$ is chosen by a prime whose vertex lies in $K$); so $F$ is a pseudoforest. For fixed $F$, the weight of the pairs $(U,f)$ with $f(U) = F$ is at most $\prod_{v \in V(F)} (1 + z \deg_F(v)) \le e^{2z|V(F)|}$. For the singles, independence gives $E[\prod_{\ell \in U_s} s_\ell \mathbf{1}[F \text{ occurs}]] \le \prod_{\ell \in U_s \setminus \pi(F)} g_\ell \cdot P(F \text{ occurs})$, $\pi(F)$ being the primes

of $V(F)$; summing $z^{|U_s|}$ over $U_s$ costs $(1+z)^{|\pi(F)|}\prod_\ell(1+zg_\ell)$. Hence

$$E\prod_\ell(1+za_\ell) \le e^{zS_1}\sum_F e^{3z|V(F)|}P(F \text{ occurs})$$
$$\le e^{zS_1}\exp\Big(\sum_K e^{3z|V(K)|}P(K \text{ occurs})\Big),$$

$K$ running over connected pseudoforests; here we used that $P(F)$ is the product over components when these lie on disjoint primes and 0 otherwise. A connected pseudoforest on $v$ vertices is a spanning tree $T$ plus at most one of $\le v^2/2$ further pairs, and $P(K)\le P(T)\le\prod_{w\in T}p(w)$. Encode a tree rooted at $v_0$ by its child sets $C_1,\dots,C_v$ in breadth-first order (children ordered by a fixed total order); every tree is recovered once for each choice of root, so summing over roots overcounts by the factor $v$. Summing over $C_i\subseteq N(v_i)$ with $|C_i|=c_i$ gives at most $\deg(v_i)^{c_i}/c_i!$, and for $i=1$, $c_1\ge1$, at most $\deg(v_0)\delta^{c_1-1}/c_1!$; and $\sum_{c_1+\cdots+c_v=v-1}\prod 1/c_i! = v^{v-1}/(v-1)!\le e^v$. So the trees with $v\ge2$ vertices contribute at most $\sum_{v_0}p(v_0)\deg(v_0)e^v\delta^{v-2} = 2S_2e^v\delta^{v-2}$, and

$$\sum_K e^{3z|V(K)|}P(K)\le 2S_2e^{6z+2}\sum_{j\ge0}(3+2j+\tfrac{j^2}{2})(e^{3z+1}\delta)^j \le 16e^{6z+2}S_2,$$

as $e^{3z+1}\delta\le e^{-1}$ and $\sum_j(3+2j+j^2/2)e^{-j}<7.6$. □

The hypothesis is a bound on *vertex degrees*; configurations in which a complete bipartite graph of events fires (the hubs of Proposition 13.6) are not excluded. It is enforced as follows.

**Lemma 8.2** (Proved)**.** *For $\delta>0$ let $H=\{v:\deg(v)>\delta\}$. Replace $S_\ell$ by $S_\ell^+ = S_\ell\cup\bigcup_{v\in H \text{ at } \ell}V_v$ and delete the edges with an endpoint in $H$. Then the absence of every new event implies the absence of every old event; every remaining vertex has degree $\le\delta$; $g_\ell^+\le g_\ell + w_\ell/\delta$, $S_1^+\le S_1+2S_2/\delta$ and $S_2^+\le S_2$.*

*Proof.* An old edge at $v\in H$ cannot occur without the new single at $\ell_v$. Degrees only decrease. By Markov, $\sum_{v\in H \text{ at } \ell}p(v)\le\sum_{v \text{ at } \ell}p(v)\deg(v)/\delta = w_\ell/\delta$, and $\sum_\ell w_\ell = 2S_2$. □

## 9. An abstract criterion

**Lemma 9.1** (Local lemma with conditioning; classical, proof included)**.** *Let $\mathcal{A}$ be finitely many events in a product probability space, each depending on a finite set of coordinates; write $A'\sim A$ if they share a coordinate. Suppose $x_A\in[0,1)$ satisfy $P(A)\le x_A\prod_{A'\sim A}(1-x_{A'})$. Then $P(\bigcap_{\mathcal{A}}\bar A)\ge\prod_{\mathcal{A}}(1-x_A)$, and for every event $B$ depending on a set of coordinates, $P(B\mid\bigcap_{\mathcal{A}}\bar A)\le P(B)\prod_{A\in\Gamma(B)}(1-x_A)^{-1}$, where $\Gamma(B)$ is the set of $A\in\mathcal{A}$ sharing a coordinate with $B$.*

*Proof.* The first statement is [1, Lemma 5.1.1], whose proof shows $P(A \mid \bigcap_{A'\in\mathcal{S}} \bar{A}') \le x_A$ for every $\mathcal{S} \subseteq \mathcal{A}$ with $A \notin \mathcal{S}$. For the second (cf. [6, Thm. 2.1]), $B$ is independent of $\bigcap_{A\notin\Gamma(B)} \bar{A}$, so

$$P\Big(B \;\Big|\; \bigcap_{\mathcal{A}} \bar{A}\Big) \le \frac{P(B)\,P(\bigcap_{A\notin\Gamma(B)} \bar{A})}{P(\bigcap_{\mathcal{A}} \bar{A})} = \frac{P(B)}{P(\bigcap_{\Gamma(B)} \bar{A} \mid \bigcap_{A\notin\Gamma(B)} \bar{A})},$$

and the chain rule with the displayed claim bounds the denominator below by $\prod_{\Gamma(B)}(1-x_A)$. □

*Congruence systems.* Fix $Q$ with $2 \mid Q$, a finite set $\mathcal{P}$ of primes coprime to $Q$, and exponents $e_\ell \ge 1$; put $X_\ell = n \bmod \ell^{e_\ell}$. The Haar measure is the uniform measure on $\prod_\ell (\mathbb{Z}/\ell^{e_\ell})^\times$; in it Theorem 5.1 computes $\mu$. A single is a set $S_\ell$ of unit classes mod $\ell^{e_\ell}$, a vertex a unit class mod $\ell$ (so $p(v) = 1/(\ell-1)$), an edge a unit class mod $\ell\ell'$. The pointwise statements of §7 hold for every integer $n$ (non-unit residues trigger no event).

**Theorem 9.2** (Proved)**.** *Let $\delta = e^{-50}$ and consider a graph-type congruence system with*

(G) $g_\ell \le 1/32$ *for all* $\ell \in \mathcal{P}$*;*
(W) $w_\ell \le \delta/32$ *for all* $\ell \in \mathcal{P}$*;*
(I) *for every integer* $n \equiv 1 \pmod{Q}$ *at which no single and no edge occurs,* $W(n) > T$*.*

*Put $\Sigma = S_1 + S_2$. Then there is a minorant $B$ as in Theorem 5.1 (moduli coprime to $Q$, unit classes, $B(n) \le \mathbf{1}[W(n) > T]$ for $n \equiv 1$ $(Q)$) whose moduli are products of at most $C(\Sigma+1)$ of the $\ell^{e_\ell}$, with $\mu \ge \exp(-C(\Sigma+1))$, $\log(M_1/\mu) \le C(\Sigma+1)$, and $|\mu_\psi| \le \mu/4$ for every real primitive $\psi$ as in Theorem 5.1. Here $C$ is absolute.*

*Proof. Hubs.* Apply Lemma 8.2 with $\delta$: by (W), $g_\ell^+ \le 1/16$, $S_1^+ \le S_1 + 2S_2/\delta$, all degrees are $\le \delta$, and (I) persists. Let $A(n)$ be the occurring events of the new system.

*Local lemma.* Apply Lemma 9.1 to all singles and edges with $x_E = 2P(E) \le 1/8$. For a single at $\ell$ the neighbours are the edges at $\ell$ and $\prod(1-x) \ge 1 - \sum x \ge 1 - 2w_\ell \ge 1/2$; for an edge at $\ell, \ell'$ the neighbours are the two singles and the edges at $\ell$ or $\ell'$, and $\prod(1-x) \ge (7/8)^2(1-2w_\ell-2w_{\ell'}) \ge 1/2$. So the hypothesis of Lemma 9.1 holds, and since $1 - x \ge e^{-1.1x}$ for $0 \le x \le 1/8$ and $\sum_E x_E = 2(S_1^+ + S_2^+) \le 2(S_1^+ + S_2)$,

$$P(A = \emptyset) \ge \prod_E (1 - x_E) \ge e^{-\lambda}, \qquad \lambda = 3(S_1^+ + S_2).$$

*Minorant.* With $\Lambda = \Lambda_{16} \le 16S_1^+ + 16e^{98}S_2$ (Lemma 8.1 with $z = 16$, $\delta = e^{-50} = e^{-3z-2}$), let $L$ be least with $4^{L+1} \ge 200e^{\Lambda+\lambda}$ and $B = B_L^*$ (Lemma 7.2), expanded into unit classes (Lemma 7.3; every event is a set of unit classes). Then $B \le \mathbf{1}[A = \emptyset] \le \mathbf{1}[W > T]$ on $n \equiv 1$ $(Q)$, by (I). By Lemma 7.3, $E|B - \mathbf{1}[A = \emptyset]| \le 2 \cdot 4^{-L-1}e^{\Lambda} \le e^{-\lambda}/100 \le P(A = \emptyset)/100$, so $\mu = E\,B \ge 0.99e^{-\lambda}$; and $M_1 \le M_1(B_L) + 4^{L+1}M_1(G_{L+1}) \le$

$e^{\Lambda} + 4^{L+1}16^{-L-1}e^{\Lambda} \le 2e^{\Lambda}$, so $\log(M_1/\mu) \le \Lambda + \lambda + 1$. Each term involves at most $2(L+1)$ primes, and $\Lambda, \lambda, L$ are $O(\Sigma + 1)$.

*Twist.* Let $\psi$ be real primitive of conductor $f > 1$, $\gcd(f, Q) = 1$, $f \mid d_i$ for some $i$. As $2 \mid Q$, $f$ is odd, hence squarefree, and its primes lie in $\mathcal{P}$; fix one, $\ell_0$. If $f \nmid d_i$, some prime of $f$ does not divide $d_i$ and the character sum over that coordinate vanishes; hence $\mu_\psi = E[B\psi]$ (Haar) and

$$|\mu_\psi| \le E|B - \mathbf{1}[A = \emptyset]| + |E[\mathbf{1}[A = \emptyset]\psi]|.$$

Let $\bar{A}'$ be the event that no event whose support avoids $\ell_0$ occurs; it depends on $X_{-\ell_0} = (X_\ell)_{\ell \ne \ell_0}$ only. Then $\mathbf{1}[A = \emptyset] = \mathbf{1}_{\bar{A}'} \cdot \mathbf{1}[X_{\ell_0} \notin \mathrm{Forb}]$, where $\mathrm{Forb} = \mathrm{Forb}(X_{-\ell_0})$ is the union of $S^+_{\ell_0}$ and of the vertices at $\ell_0$ joined to an occurring vertex. Write $\psi = \chi_0(X_{\ell_0})\psi'(X_{-\ell_0})$ with $\chi_0 = (\cdot/\ell_0)$. Conditioning on $X_{-\ell_0}$ and using independence and $E\chi_0(X_{\ell_0}) = 0$,

$$\big|E_{X_{\ell_0}}[\mathbf{1}[X_{\ell_0} \notin \mathrm{Forb}]\chi_0]\big| = \big|E_{X_{\ell_0}}[\mathbf{1}[X_{\ell_0} \in \mathrm{Forb}]\chi_0]\big| \le P_{X_{\ell_0}}(\mathrm{Forb}),$$

so $|E[\mathbf{1}[A = \emptyset]\psi]| \le P(\bar{A}' \cap \{X_{\ell_0} \in \mathrm{Forb}\})$, which is at most

$$P(\bar{A}')\Big(g^+_{\ell_0} + \sum_{\{u,w\},\, u \text{ at } \ell_0} p(u)P(w \mid \bar{A}')\Big) \le P(\bar{A}')\big(\tfrac{1}{16} + 1.22 w_{\ell_0}\big) < 0.064\, P(\bar{A}').$$

Here the single at $\ell_0$ and $X_{\ell_0}$ are independent of $\bar{A}'$ and of $X_{\ell_w}$, and Lemma 9.1, applied to the family defining $\bar{A}'$ and $B = \{X_{\ell_w} \in V_w\}$ (neighbours: the single and the edges at $\ell_w$, with $\prod(1-x) \ge (7/8)(15/16) > 0.82$), gives $P(w \mid \bar{A}') \le 1.22 p(w)$. As $P(A = \emptyset) = P(\bar{A}') - P(\bar{A}' \cap \{X_{\ell_0} \in \mathrm{Forb}\}) \ge 0.936 P(\bar{A}')$, we get $|\mu_\psi| < 0.08 P(A = \emptyset) < \mu/4$. □

## 10. Private covers and a hypergraph moment bound

*Hypergraph-type systems.* As in §8, with singles $S_\ell$ and vertices $v = (\ell, V_v)$ (vertices at one prime pairwise disjoint), but with *hyperedges* in place of edges: a hyperedge is a set of $2, \dots, k$ vertices at distinct primes, occurring iff all its vertices are realised, so $P(e) = \prod_{v \in e} p(v)$. Repeated hyperedges are allowed (they only enlarge the quantities below). Put $S_H = \sum_e P(e)$, $w_\ell = \sum_{e \text{ at } \ell} P(e)$, and for a set $O$ of vertices at distinct primes the *codegree*

$$\Delta_O = \sum_{e \supsetneq O} \prod_{v \in e \setminus O} p(v), \qquad \Delta^{(i)} = \max_{|O| = i} \Delta_O;$$

thus $\Delta_{\{v\}} = \deg(v)$.

For a set $P$ of primes, a *private cover* of $P$ is a set $\mathcal{C}$ of events with $P \subseteq \bigcup_{E \in \mathcal{C}} \operatorname{supp} E$ such that every $E \in \mathcal{C}$ has a prime of $\operatorname{supp} E \cap P$ lying in the support of no other member of $\mathcal{C}$. A family is *private* if each member has a prime lying in no other member. Put $\pi(\mathcal{C}) = \bigcup_{E \in \mathcal{C}} \operatorname{supp} E$ and

$$G^{\mathrm{cov}}_u(x) = \sum_{|P| = u} \sum_{\mathcal{C} \text{ private cover of } P} \mathbf{1}[\text{every } E \in \mathcal{C} \text{ occurs at } x].$$

**Lemma 10.1** (Proved). *For every $x$ and $u, L \geq 0$: $\binom{N(x)}{u} \leq G_u^{\mathrm{cov}}(x)$; hence Lemma 7.2 holds with $G_{L+1}^{\mathrm{cov}}$ in place of $G_{L+1}$. If every event is a union of cells, then $M_1(B_L) \leq \sum_{u \leq L} 2^u E\, G_u^{\mathrm{cov}}$ and $M_1(G_u^{\mathrm{cov}}) = E\, G_u^{\mathrm{cov}}$.*

*Proof.* Every $P \subseteq V(x)$ is covered by occurring events, and a minimal covering subfamily is a private cover (a member without a private prime could be removed). This gives the first claim, and the proof of Lemma 7.2 goes through verbatim. In the proof of Lemma 7.3, a cell $c$ on $U$ with $\kappa(U, c) \neq 0$ has every prime of $U$ covered by events supported in $U$ occurring on $c$; a minimal such subfamily is a private cover of $U$, so $\sum_c |\kappa(U, c)| P(c) \leq 2^{|U|} \sum_{\mathcal{C} \text{ priv. cover of } U} P(\mathcal{C} \text{ occurs})$. □

**Lemma 10.2** (Proved). *Let $U_0 \geq 1$, $\rho \geq 1$ and put $D = \sum_{j=0}^{k-2} (kU_0)^j \Delta^{(j+1)}$. If $D \leq [2ek\rho^k]^{-1}$, then*

$$\sum_{\mathcal{C} \text{ private}, \ |\mathcal{C}| \leq U_0} \rho^{|\pi(\mathcal{C})|} P(\mathcal{C} \text{ occurs}) \leq \exp\left(\rho S_1 + 2ek\rho^k S_H\right).$$

*In particular, for $w \geq 0$ and $\rho = 1 + w$, $\sum_{u \leq U_0} w^u E\, G_u^{\mathrm{cov}} \leq \exp((1+w)S_1 + 2ek(1+w)^k S_H)$.*

*Proof.* The last claim follows from the first, because a private family privately covers only subsets of $\pi(\mathcal{C})$, so pointwise $\sum_{u \leq U_0} w^u G_u^{\mathrm{cov}} \leq \sum_{\mathcal{C} \text{ private}, |\mathcal{C}| \leq U_0} (1+w)^{|\pi(\mathcal{C})|} \mathbf{1}[\mathcal{C} \text{ occurs}]$.

A single in a private family has its only prime private, so the singles sit on primes disjoint from the rest and contribute at most $\prod_\ell (1 + \rho g_\ell) \leq e^{\rho S_1}$. The hyperedge part splits into connected components, each private; two components using the same prime with different vertices cannot occur together, and otherwise $P$ and $\rho^{|\pi|}$ are multiplicative over components. So the hyperedge part is at most $\exp(\sum_K \rho^{|V(K)|} P(K))$ over connected private $K$ with $h = |K| \leq U_0$ hyperedges, $|V(K)| \leq kh$.

*Exploration.* Pick a first hyperedge $e_1 \in K$ and declare its vertices discovered; process discovered vertices in breadth-first order (ties by a fixed order), attaching at a vertex $v$ as *children* all not-yet-attached hyperedges of $K$ containing $v$. Every hyperedge is attached ($K$ is connected), and a child $e$ of $v$ has a private vertex, which is therefore new. If $O_e$ denotes $v$ together with the old vertices of $e$, $|O_e| = j + 1$, the weight of $e$ beyond $O_e$ is the product of $p$ over its new vertices, and the sum over admissible $e$ is at most $\sum_j \binom{kh}{j} \Delta^{(j+1)} \leq D$. To make the encoding independent of $K$, record the child sets in $N := kh$ fixed breadth-first slots, padding unused slots with empty sets; $K \mapsto (e_1, \text{child sets})$ is injective.

*Counting.* The first hyperedge contributes $\sum_{e_1} P(e_1) \cdot \#\text{roots} \leq kS_H$ (the root index only overcounts). A slot with $c$ children contributes at most $D^c/c!$: the product of the sibling weights is $\prod p(u)$ over all vertices they bring, independent of order, so the sum over sets is $1/c!$ times the sum over ordered tuples, which is at most $D^c$ since each next child has a private, hence new, vertex and its old vertices lie in a pool of at most $kh$ vertices. The number of

weighted compositions is $\sum_{c_1+\cdots+c_N=h-1}\prod 1/c_i! = N^{h-1}/(h-1)! \le e(ke)^{h-1}$. Hence $\sum_{|K|=h}\rho^{|V(K)|}P(K) \le \rho^{kh}\, e\, kS_H\,(keD)^{h-1}$, and summing over $h \ge 1$ with $\rho^k keD \le 1/2$ gives at most $2ek\rho^k S_H$. □

The factor $(kU_0)^j$ in $D$ is real: a cluster of hyperedges reusing $j+1$ old vertices out of many costs only $\Delta^{(j+1)}$ per hyperedge but can be placed in many ways. For the multiplier problem some pair codegrees are of constant size (Remark 11.6), so Lemma 10.2 does not apply directly to the whole system with $U_0$ of the size of the total mass, and (*Assessment*) quarantining the heavy pairs inside the same truncation does not seem affordable; §11 circumvents this.

## 11. Multilevel minorants

*Levels.* Consider a congruence system as in §9 whose events are singles and hyperedges of $2,\dots,k$ vertices at distinct primes, all vertices being unit classes mod $\ell^{e_\ell}$. Its events are partitioned into *levels*: level 2 consists of the singles and the edges, and level $r$ ($3 \le r \le k$) of hyperedges with $r$ vertices. Let $F_r$ be the indicator that no event of level $r$ occurs, $F_{<r} = \prod_{2\le s<r} F_s$, $\mathcal{A}_{<r} = \{F_{<r} = 1\}$, and let $\Sigma_r$ be the mass of level $r$ ($\Sigma_2 = S_1 + S_2$). We assume throughout

($\mathrm{P}_k$) the total mass of all events at every prime is at most $1/(64k)$.

**Lemma 11.1** (Proved)**.** *Under* ($\mathrm{P}_k$)*, with* $x_E = 2P(E)$*:*

(a) $P(\mathcal{A}_{<r+1}) \ge P(\mathcal{A}_{<r})\prod_{E\in \text{level } r}(1-x_E) \ge P(\mathcal{A}_{<r})e^{-3\Sigma_r}$ *for every* $r$*, and* $P(\mathcal{A}_{<3}) \ge e^{-3\Sigma_2}$*;*

(b) *for every* $r$ *and every event* $B$ *depending on the coordinates in a set* $U$ *of primes,* $P(B\cap\mathcal{A}_{<r}) \le P(\mathcal{A}_{<r})P(B)e^{|U|/2}$*.*

*Proof.* An event meets at most $k$ primes, so the $x$-sum over its neighbours is at most $k\cdot 2/(64k) = 1/32$ and the hypothesis of Lemma 9.1 holds for the union of all levels. Its proof gives $P(E \mid \bigcap_{E'\in\mathcal{S}}\bar{E}') \le x_E$ for all $\mathcal{S}$; the chain rule over the events of level $r$, conditioned on $\mathcal{A}_{<r}$ and the earlier ones, gives (a), using $1-x \ge e^{-1.1x}$. For (b) apply the second part of Lemma 9.1 to the levels $<r$: $\sum_{A\in\Gamma(B)} x_A \le |U|/32$, so $\prod_{\Gamma(B)}(1-x_A)^{-1} \le e^{1.2|U|/32} \le e^{|U|/2}$. □

*Constants.* $z = 16$, $\delta = e^{-50}$ (level 2, as in Theorem 9.2); $w = 16$, $\rho = 17e^{1/2}$ and $\delta_r = [4er\rho^r]^{-1}$ (levels $\ge 3$). For $\hat{S} \ge 1$ put $\Lambda_3 = 6e\rho^3\hat{S}$, let $L_3$ be least with $4^{L_3+1} \ge 400e^{\Lambda_3+3\hat{S}}$, and $t(\hat{S}) = \delta_3/(3(L_3+1))$.

**Theorem 11.2** (Two-level minorant; proved)**.** *Take a system with levels* 2 *and* 3 *only, satisfying* ($\mathrm{P}_3$)*, and*

(D2) *every level-*2 *vertex has level-*2 *degree* $\le \delta$*;*

(D3) *for some* $\hat{S} \ge \Sigma_3$*, every level-*3 *vertex has degree* $\le \delta_3$ *and every pair of level-*3 *vertices has codegree* $\le t(\hat{S})$*.*

*Put $\Sigma = \Sigma_2 + \hat{S}$. There is a real combination $B$ of unit cells with $B \le F_2F_3$ pointwise, $E[F_2F_3 - B] \le P(\mathcal{A}_{<4})/100$ (so $\mu = E\,B \ge 0.99e^{-3\Sigma}$), $\log(M_1/\mu) \le C(\Sigma+1)$, every cell on at most $C(\Sigma+1)$ primes, and $|E[B\psi]| < \mu/4$ for every real primitive $\psi$ as in Theorem 5.1. Here $C$ is absolute.*

*Proof. Level* 3. For the level-3 events let

$$B_3 = B_{L_3} - 4^{L_3+1}G^{\mathrm{cov}}_{L_3+1} \qquad \text{(Lemma 10.1)};$$

pointwise $B_3 \le F_3$ and $|F_3 - B_3| \le 2 \cdot 4^{L_3+1}G^{\mathrm{cov}}_{L_3+1}$. Multiply the pointwise inequality in the proof of Lemma 10.2 by $F_2$ and take expectations; Lemma 11.1(b) with $U = \pi(\mathcal{C})$ gives

$$E[F_2\mathbf{1}[\mathcal{C}]] \le P(\mathcal{A}_{<3})P(\mathcal{C})e^{|\pi(\mathcal{C})|/2},$$

and $(1+w)e^{1/2} = \rho$. By (D3), $D = \deg + 3(L_3+1)\Delta^{(2)} \le 2\delta_3$, which is $[2e\cdot 3\rho^3]^{-1}$, so Lemma 10.2 (with $U_0 = L_3+1$, no singles) gives $E[F_2G^{\mathrm{cov}}_{L_3+1}] \le 16^{-L_3-1}P(\mathcal{A}_{<3})e^{\Lambda_3}$ and therefore

$$E[F_2(F_3 - B_3)] \le 2 \cdot 4^{-L_3-1}P(\mathcal{A}_{<3})e^{\Lambda_3} \le P(\mathcal{A}_{<3})e^{-3\hat{S}}/200 \le P(\mathcal{A}_{<4})/200$$

by Lemma 11.1(a). Without the tilt, $M_1(B_3) \le 2e^{\Lambda_3}$ by Lemmas 10.1 and 10.2, and every cell of $B_3$ lies on at most $3(L_3+1)$ primes.

*Level* 2, *cell by cell.* Write $B_3 = \sum_i c_i\mathbf{1}_{\mathcal{C}_i}$ with cells $\mathcal{C}_i$ on prime sets $P_i$. If $\mathcal{C}_i$ realises a level-2 single at a prime of $P_i$, or both ends of a level-2 edge inside $P_i$, then $F_2 = 0$ on $\mathcal{C}_i$; put $\beta_i = \alpha_i = 0$. Otherwise, on $\mathcal{C}_i$, $F_2$ equals the void indicator $F_2^{(i)}$ of the *conditioned* level-2 system on $\mathcal{P} \setminus P_i$: the original singles and edges there, plus the induced singles $\{w : \{v,w\}$ an edge, $v$ realised by $\mathcal{C}_i\}$, of total mass $\le \delta|P_i|$; degrees remain $\le \delta$. Lemmas 7.2, 7.3 and 8.1 give, with $\Lambda_2 = 16(S_1 + 3\delta(L_3+1)) + 16e^{98}S_2$ and every $L$, functions $B_L^{(i)}$, $G_{L+1}^{(i)}$ with $|B_L^{(i)} - F_2^{(i)}| \le 4^{L+1}G_{L+1}^{(i)}$, $E\,G_{L+1}^{(i)} \le 16^{-L-1}e^{\Lambda_2}$ and $M_1(B_L^{(i)}) \le e^{\Lambda_2}$. Let $L_2$ be least with $4^{L_2+1} \ge 800e^{\Lambda_2+\Lambda_3+3\Sigma}$, put $\beta_i, \alpha_i = B_{L_2}^{(i)} \mp 4^{L_2+1}G_{L_2+1}^{(i)}$ (so $\beta_i \le F_2^{(i)} \le \alpha_i$), and

$$B = \sum_{c_i>0} c_i\beta_i\mathbf{1}_{\mathcal{C}_i} - \sum_{c_i<0} |c_i|\alpha_i\mathbf{1}_{\mathcal{C}_i}.$$

Then $B \le \sum_i c_iF_2\mathbf{1}_{\mathcal{C}_i} = F_2B_3 \le F_2F_3$. As $\beta_i, \alpha_i$ do not depend on the coordinates in $P_i$,

$$E[F_2F_3 - B] \le E[F_2(F_3 - B_3)] + \sum_i |c_i|P(\mathcal{C}_i)\,2 \cdot 4^{-L_2-1}e^{\Lambda_2}$$

$$\le \frac{P(\mathcal{A}_{<4})}{200} + \frac{e^{-3\Sigma}}{200} \le \frac{P(\mathcal{A}_{<4})}{100},$$

since $P(\mathcal{A}_{<4}) \ge e^{-3\Sigma}$. Further $M_1(B) \le M_1(B_3) \cdot 2e^{\Lambda_2} \le 4e^{\Lambda_2+\Lambda_3}$, every cell lies on at most $3(L_3+1) + 2(L_2+1)$ primes, and $L_3, \Lambda_3 = O(\hat{S}+1)$, $\Lambda_2, L_2 = O(\Sigma+1)$.

*Twist.* As in Theorem 9.2, $\mu_\psi = E[B\psi]$ and $|E[B\psi]| \le E[F-B] + |E[F\psi]|$ with $F = F_2F_3$. For a prime $\ell_0 \mid f$ let $\bar{A}'$ be the event that no event (of either

level) avoiding $\ell_0$ occurs. Then $|E[F\psi]| \le P(\bar{A}' \cap \{\text{an event at } \ell_0 \text{ occurs}\})$, and by Lemma 9.1 applied to the family defining $\bar{A}'$ (the rest of an event through $\ell_0$ meets at most two further primes, with neighbour product $\ge (1-1/96)^2$) this is at most $1.1 \cdot P(\bar{A}')/192 < 0.01P(\bar{A}')$. Hence $|\mu_\psi| < 0.03P(\mathcal{A}_{<4}) < \mu/4$. □

*Pushing heavy sets down.* Theorem 11.2 assumes small codegrees only for level 3, and only at the scale $t(\hat{S})$, which depends on $\hat{S}$ alone. Heavy pairs are moved into level 2, whose own truncation is unconstrained.

**Lemma 11.3** (Proved)**.** *Start from one system of singles, edges and 3-vertex hyperedges with masses $S_1, S_2, S_H$, in which every prime carries total mass $\le c_0$. Let $\hat{S} \ge S_H$, $t = t(\hat{S})$, and $c_0 \le \delta\delta_3 t/1728$. Perform in order:* (a) *every vertex of hyperedge-degree $> \delta_3$ becomes a single and the hyperedges through it are deleted;* (b) *every vertex pair $O$ with codegree $\Delta_O > t$ becomes an edge (merged with an existing edge if any) and the hyperedges containing it are deleted;* (c) *every vertex of edge-degree $> \delta$ becomes a single and the edges and hyperedges through it are deleted. The result is a two-level system satisfying* (P$_3$), (D2) *and* (D3) *with this $\hat{S}$; if no event of it occurs, no original event occurs; and with $S_2^{(b)}$ the edge mass after* (b)*,*

$$\Sigma_3 \le S_H, \quad S_2^{(b)} \le S_2 + 3S_H/t, \quad S_1^{\text{new}} \le S_1 + 3S_H/\delta_3 + 2S_2^{(b)}/\delta.$$

*Proof.* Deletions only lower degrees and codegrees, so after (c) the conditions (D2), (D3) hold (pushed pairs have codegree 0). Every deleted event contains a new single or edge, so it cannot occur when no new event occurs. Markov's inequality, per prime: (a) adds at most $w_\ell^{(3)}/\delta_3$, since $\sum_{v \text{ at } \ell} p(v)\deg(v) = w_\ell^{(3)}$; (b) adds at most $\sum_{O \ni v \text{ at } \ell} P(O)\Delta_O/t = 2w_\ell^{(3)}/t$; (c) adds at most $w_\ell^{(2),(b)}/\delta$. So the new per-prime total is at most $c_0(2 + 1/\delta_3 + (1+2/t)(1+1/\delta)) \le 9c_0/(t\delta\delta_3) \le 1/192$, which is (P$_3$). The global bounds are the same sums over $\ell$, with $\sum_\ell w_\ell^{(3)} = 3S_H$ and $\sum_\ell w_\ell^{(2)} = 2S_2$. □

**Corollary 11.4** (Proved)**.** *Consider a congruence system of singles, edges and 3-vertex hyperedges satisfying* (I) *of Theorem 9.2, with $\hat{S} \ge S_1 + S_2 + S_H$ and every prime carrying total mass $\le c_0(\hat{S}) := \delta\delta_3 t(\hat{S})/1728$. Then there is a minorant $B$ as in Theorem 5.1, with the twist condition, such that $\log(M_1/\mu)$ and the number of prime powers in each modulus are $\le C(\hat{S}^2 + 1)$.*

*Proof.* Lemma 11.3, then Theorem 11.2 with $\Sigma \le S_1 + 3\hat{S}/\delta_3 + (1+2/\delta)S_2^{(b)} + \hat{S} = O(\hat{S}/t) = O(\hat{S}^2 + 1)$; (I) transfers since $B \le F_2F_3 \le \mathbf{1}[\text{no original event}]$. □

**Theorem 11.5** ($k$-level minorant; proved)**.** *For every $k \ge 3$ there are constants $C_k, A_k$ with the following property. Consider a congruence system of singles and hyperedges with $2, \dots, k$ vertices satisfying* (I) *of Theorem 9.2, with total mass $\le \hat{S}$ ($\hat{S} \ge 1$) and every prime carrying total mass $\le c_k(\hat{S}) := \hat{S}^{-A_k}/C_k$. Then there is a minorant $B$ as in Theorem 5.1, with the twist*

*condition, such that* $\log(M_1/\mu)$ *and the number of prime powers in each modulus are at most* $C_k\hat{S}^{A_k}$.

*Proof.* Put events with $r$ vertices into level $r$ ($3 \le r \le k$) and singles and edges into level 2.

*Step A (push-down, top to bottom).* For $r = k, k-1, \dots, 3$ in this order, let $\Sigma_r$ be the mass of level $r$ at the moment it is processed (it contains the sets pushed into it from above; later steps only delete events of level $\ge r$ or add events below, so $\Sigma_r$ stays an upper bound), and put

$$H_r = k\sum_{s>r}(L_s+1), \qquad \hat{S}_{\ge r} = \sum_{s\ge r}(\Sigma_s+1),$$

$$\Lambda_r = 2er\rho^r(\Sigma_r + H_r + 1), \qquad \mathfrak{M}_{r+1} = \prod_{s>r} 4e^{\Lambda_s},$$

$$L_r = \min\{L:\ 4^{L+1} \ge 200k\,\mathfrak{M}_{r+1}e^{H_r/2+\Lambda_r+3\hat{S}_{\ge r}}\},$$

$$N_r = r(L_r+1) + H_r.$$

Then: (a) every vertex of level-$r$ degree $> \delta_r/2$ becomes a single; (b) for $1 \le j \le r-2$, every vertex set $O$ with $|O| = j+1$ and $\Delta_O^{(r)} > \delta_r(4N_r)^{-j}/(2r)$ becomes an event of level $j+1$; every level-$r$ event containing a pushed vertex or set is deleted. Finally, level-2 vertices of degree $> \delta$ become singles, deleting the events through them. As in Lemma 11.3, no deleted event can occur when no remaining event occurs, and by Markov a set pushed from level $r$ carries per-prime mass $\le 2^r w_\ell^{(r)}(4N_r)^j\,2r/\delta_r$ and total mass $\le 2^r\Sigma_r(4N_r)^r\,2r/\delta_r$.

*The induction.* (i) $\log\mathfrak{M}_{r+1} = \sum_{s>r}(\log 4 + \Lambda_s)$, and $\Lambda_r$, $H_r$, $\hat{S}_{\ge r}$ are linear in $\Sigma_r$ and in the earlier quantities $L_s, \Sigma_s$ ($s > r$); hence $L_r, N_r = O_k(1 + \sum_{s\ge r}\Sigma_s + H_r)$, and inductively $L_r, N_r, H_r \le C\hat{S}^A$ once $\Sigma_s \le C\hat{S}^A$ for $s \ge r$, with $C, A$ depending on $k$ only. (ii) The pushes from level $r$ add total mass $\le 2^r\Sigma_r(4N_r)^r\,2r/\delta_r$ to the lower levels, and at every prime they add at most $2^r(4N_r)^r\,2r/\delta_r$ times the per-prime mass of level $r$; level-2 hub vertices add at most $1/\delta$ times the level-2 per-prime mass. Starting from $\Sigma_k \le \hat{S}$ and per-prime masses $\le c_k(\hat{S})$, after the $k-2$ steps every $\Sigma_r$ and $L_r, N_r, H_r$ is $\le C'_k\hat{S}^{A'_k}$, and every per-prime mass is $\le c_k(\hat{S})\,C'_k\hat{S}^{A'_k}$, where $C'_k, A'_k$ depend on $k$ only (not on $c_k$). (iii) Taking $A_k \ge A'_k$ and $C_k \ge 64k\,C'_k$, the per-prime total is $\le \hat{S}^{A'_k - A_k}C'_k/C_k \le 1/(64k)$, which is ($\mathrm{P}_k$).

*Step B (composition, top level innermost).* Let $B_{\ge k}$ be the minorant $B_{L_k} - 4^{L_k+1}G^{\mathrm{cov}}_{L_k+1}$ of $F_k$. Given $B_{\ge r+1} = \sum_i c_i\mathbf{1}_{\mathcal{C}_i}$ with cells $\mathcal{C}_i$ on $P_i$, $|P_i| \le H_r$, form the level-$r$ system conditioned on $\mathcal{C}_i$: events meeting $P_i$ are realised, vanish, or lose their realised vertices, the last becoming *induced* events of smaller support; identical induced events are merged (this changes neither the void indicator nor any bound below). If an event is fully realised, put $\beta_i = \alpha_i = 0$; otherwise let $\beta_i, \alpha_i = B_{L_r} \mp 4^{L_r+1}G^{\mathrm{cov}}_{L_r+1}$ for the conditioned void indicator $F_r^{(i)}$, and put $B_{\ge r} = \sum_{c_i>0} c_i\beta_i\mathbf{1}_{\mathcal{C}_i} - \sum_{c_i<0}|c_i|\alpha_i\mathbf{1}_{\mathcal{C}_i}$. Level 2

is treated in the same way with Lemmas 7.2–8.1, at the least $L_2$ with $4^{L_2+1} \geq 200k\,\mathfrak{M}_3 e^{\Lambda_2+3\hat{S}_{\geq 2}}$, where $\Lambda_2 = 16(S_1+\delta H_2)+16e^{98}S_2$ (induced singles have mass $\leq \delta H_2$). Finally $B = B_{\geq 2}$.

*Conditioned codegrees.* An induced event $e \setminus F'$ ($F'$ its realised fixed vertices) contains $O$ iff $e \supseteq O \cup F'$; there are at most $H_r \leq N_r$ fixed vertices. By (a) and (b), and $H_r/(4N_r) \leq 1/4$,

$$\Delta'_v \leq \frac{\delta_r}{2} + \sum_{i\geq 1} \frac{H_r^i \delta_r}{2r(4N_r)^i} \leq \delta_r,$$

$$\Delta'_O \leq \sum_{i\geq 0} \frac{H_r^i \delta_r}{2r(4N_r)^{|O|+i-1}} \leq \frac{\delta_r}{r(4N_r)^{|O|-1}} \quad (|O|\geq 2).$$

As $r(L_r+1) \leq N_r$, Lemma 10.2 applies with $U_0 = L_r+1$: $\sum_j (rU_0)^j \Delta'^{(j+1)} \leq \delta_r + \sum_{j\geq 1} 4^{-j}\delta_r/r \leq 2\delta_r = [2er\rho^r]^{-1}$. The induced events have mass at most $\sum_{i=1}^{r-1}\binom{h}{i}\max_{|F'|=i}\Delta^{(r)}_{F'} \leq h\delta_r/2 + \frac{\delta_r}{2r}\sum_{i\geq 2} h^i(4N_r)^{1-i} \leq h\delta_r$ with $h \leq H_r$ fixed vertices; so the conditioned masses are $\leq \Sigma_r + H_r$, and the tilted moment is $\leq e^{\Lambda_r}$.

*Errors.* With the true lower indicators,

$$E[F_{<r}B_{\geq r}] \geq E[F_{<r}F_rB_{\geq r+1}] - \sum_i |c_i|\,E\big[F_{<r}\mathbf{1}_{\mathcal{C}_i}|\beta_i/\alpha_i - F_r^{(i)}|\big],$$

and by Lemma 11.1(b), applied to $\mathcal{C}_i \cap \{\mathcal{C} \text{ occurs}\}$ on $P_i \cup \pi(\mathcal{C})$, the last expectation is at most $2\cdot 4^{L_r+1}P(\mathcal{A}_{<r})P(\mathcal{C}_i)e^{|P_i|/2}16^{-L_r-1}e^{\Lambda_r}$. Since $M_1(B_{\geq r+1}) \leq \mathfrak{M}_{r+1}$ (each level multiplies the mass by $\leq 4e^{\Lambda_s}$), the choice of $L_r$ and Lemma 11.1(a) bound the sum by $P(\mathcal{A}_{<r})e^{-3\hat{S}_{\geq r}}/(100k) \leq P(\text{all})/(100k)$, where $P(\text{all}) = E\prod_r F_r$; level 2 contributes the same amount in Haar measure. Telescoping from $r=2$ to $r=k$ gives $B \leq \prod_r F_r$ pointwise and $E[\prod_r F_r - B] \leq P(\text{all})/100$, so $\mu \geq 0.99P(\text{all}) \geq 0.99e^{-3\hat{S}_{\geq 2}}$ and $M_1(B) \leq \mathfrak{M}_2$. The twist is handled as in Theorem 11.2: the rest of an event through $\ell_0$ meets at most $k-1$ further primes, and ($\mathrm{P}_k$) makes the conditional factor $\leq 1.1$. All quantities are $\leq C_k\hat{S}^{A_k}$. □

**Remark 11.6** (Why levels are needed; proved for the stated lower bound)**.** Every atom is $(M,D) = (4sab-1, sa^2)$ with $s$ squarefree, and its class is $-4D \equiv -a/b \equiv -4sa^2 \equiv -1/(4sb^2) \pmod M$. Fixing two of $s,a,b$ and letting the third run over a progression gives a class which is the same rational at every $M$ in a progression modulo $4sa$, $4sb$ or $4ab$. For $y = T^\theta$, $\theta < 1/3$, and distinct primes $\ell_1,\ell_2 \in (y, T^{1/3}]$, the prime number theorem in progressions shows that such a pair of vertices has codegree $\geq (\log(1/(3\theta)) - o(1))/\varphi(4n)$, $n = sa, sb, ab$ respectively (take $M = \ell_1\ell_2\ell_3$ with $\ell_3 \in (y, T^{1/3}]$ in one class mod $4n$). *Assessment* (not proved, and not used): the number of such classes with $\varphi(4n) \leq X$ should grow like $X\log X$, so that codegrees $\leq 1/L$ could not be enforced inside one truncation of level $L \asymp$ total mass, removing the heavy pairs costing more than $L$. In Theorem 11.5 these pairs become level-2

edges, their vertices have level-2 degree $\asymp 1$ and become singles, and their cost enters only $L_2$.

## 12. Proof of Theorem 1.1

For a set $\Pi$ of primes and $M \le T$, $M \equiv 3 \pmod 4$, write $M = m_\Pi r_\Pi$ with $m_\Pi$ the $\Pi$-part. An atom $(M, D)$, $D \mid A_M^2$, *survives* $\Pi$ if $m_\Pi \mid 4D+1$ and $r_\Pi > 1$; its Haar weight is $1/\varphi(r_\Pi)$. Put

$$S^* = \sum_{(M,D)} \max_{\Pi} \frac{\mathbf{1}[(M,D) \text{ survives } \Pi]}{\varphi(r_\Pi)},$$

the sum over all atoms with $M \le T$ and the maximum over all sets $\Pi$ of primes.

**Lemma 12.1** (Proved)**.** $S_{\mathrm{tot}}(\Pi) := \sum_{\text{atoms surviving } \Pi} 1/\varphi(r_\Pi) \le S^* \le \exp(O(\log T/\log\log T))$ *for every* $\Pi$.

*Proof.* The first inequality is trivial. If $(M, D)$ survives $\Pi$ then $m_\Pi$ divides $g = \gcd(M, 4D+1)$, so $1/\varphi(r_\Pi) \ll \log\log T \cdot g/M$. Follow the proof of Lemma 3.3 with $m = g$: the involution $D \mapsto A^2/D$ preserves $g$; writing $D = sr'^2$, $A = sr'k$ with $r' \le k \le X$ gives $g \mid r' + k$; for fixed $(s, r')$ each $k$ determines $M$, $D$ and hence $g$, and $g \mid 4sr'^2 + 1$. So $S^* \ll \log\log T\,(3 + \mathcal{L}) \sum_{sr'^2 \le X} \tau(4sr'^2+1)/(sr')$. □

**Lemma 12.2** (Iterated quarantine; proved)**.** *Let* $2 \le z \le T$, *put* $\kappa = \lfloor \log T/\log z \rfloor$, *let* $0 < c_0 \le 1$, *and let* $w_\ell(\Pi)$ *be the Haar mass of the distinct events surviving* $\Pi$ *whose rough part is divisible by* $\ell$. *Starting from* $\Pi_0 = \{\ell \le z\}$, *repeatedly add to* $\Pi$ *every prime* $\ell > z$ *with* $\ell \notin \Pi$ *and* $w_\ell(\Pi) > c_0$. *The process stops at some* $\Pi = \Pi_0 \cup \mathcal{B}$ *with* $|\mathcal{B}| \le \kappa S^*/c_0$, *and then every prime* $\ell > z$ *outside* $\Pi$ *has* $w_\ell(\Pi) \le c_0$.

*Proof.* If $\ell$ is added at some stage $\Pi_i$, then $c_0$ is less than the sum of $1/\varphi(r_{\Pi_i})$ over atoms with $\ell \mid r_{\Pi_i}$, and each term is at most the atom's contribution to $S^*$. Summing over the added primes, an atom is counted at most once for each of its at most $\kappa$ prime factors $> z$. □

**Proposition 12.3** (Proved)**.** *Let* $k \ge 3$ *and* $y = T^{1/k}\exp(2\mathcal{L}/\log\mathcal{L})$. *For all large* $T$ *there are* $Q$ *with* $840 \mid Q$ *and* $\log Q \le (k+1)y$, *and a minorant* $B$ *as in Theorem 5.1 with* $\mu > 0$, *the twist condition, and* $\log\max d_i$, $\log(M_1/\mu)$ *both at most* $\exp(O_k(\mathcal{L}/\log\mathcal{L}))$. *In particular* $H_{\min}(\theta)$ (*Hypothesis 13.4*) *holds for every* $\theta > 0$.

*Proof.* Put $k' = \max(k-1, 3)$, $\hat S = S^* + 1$ and $c_0 = c_{k'}(\hat S)$ (Theorem 11.5); then $1/c_0 = \hat S^{O_k(1)}$. Apply Lemma 12.2 with $z = y$; as $y^k > T$, every $M \le T$ has at most $k-1$ prime factors $> y$, so $\kappa \le k-1$ and $|\mathcal{B}| \le (k-1)S^*/c_0 = \hat S^{O_k(1)}$. Put $Q = \mathrm{lcm}(24,\ \ell^{e_\ell} :\ \ell \in \Pi)$ ($e_\ell$ maximal with $\ell^{e_\ell} \le T$) and $\mathcal{P} = \{y < \ell \le T\} \setminus \mathcal{B}$. Every atom surviving $\Pi$ has rough part $r_\Pi \le T$ with all prime factors in $\mathcal{P}$, at most $k-1$ of them counted with multiplicity; it

gives the event $n \equiv -4D \pmod{r_\Pi}$, which we write, after lifting each class mod $\ell^a$ to its classes mod $\ell^{e_\ell}$, as a union of singles or hyperedges on at most $k-1 \le k'$ vertices; identical events are merged.

*(I).* Let $n \equiv 1 \pmod{Q}$ and $M \le T$, $M \equiv 3 \pmod 4$. Every $\ell^v \| M$ with $\ell \in \Pi$ has $\ell^v \le T$, so $m_\Pi \mid Q$. If $r_\Pi = 1$, Lemma 2.2 applies. If $n \equiv -4D \pmod M$ with $D \mid A_M^2$, then $m_\Pi \mid 4D+1$, and reducing mod $r_\Pi$ shows that an event occurs. Conclude with Lemma 2.1.

*Masses.* Lifting preserves mass and merging lowers it, so every free prime carries total mass $\le w_\ell(\Pi) \le c_0$, and the total mass is $\le S_{\mathrm{tot}}(\Pi) \le S^* < \hat S$ (Lemma 12.1). Finally $\log Q \le (\pi(y) + |\mathcal{B}|)\mathcal{L} + \log 24 \le (k+1)y$, since $\mathcal{L}/\log y \to k$.

Theorem 11.5 with $k'$ now gives $B$ with $K = 1 + \log(M_1/\mu)$ and $\log \max d_i$ both at most $C_{k'}\hat S^{A_{k'}}\mathcal{L}$, which is $\exp(O_k(\mathcal{L}/\log\mathcal{L}))$ by Lemma 12.1. As $\exp(O_k(\mathcal{L}/\log\mathcal{L})) = T^{o(1)}$, the conditions of Hypothesis 13.4 hold for every $\theta > 1/k$, and $k$ is arbitrary. □

*Proof of Theorem 1.1.* Fix $k \ge 3$ and take $Q$, $B$, $y$ from Proposition 12.3; put $K = 1 + \log(M_1/\mu)$. As in Theorem 13.5, adjoin to $Q$ a prime $\ell_0 \in (R, 2R]$, $R = \max(T, \max d_i)$; then $\log Z \le (k+2)y$ and Theorem 5.1 gives a prime $p \equiv 1 \pmod{Q\ell_0}$ with $W(p) > T$ and $\log p \le C_1 K \max(\log Z, K) \le T^{1/k}\exp(O_k(\mathcal{L}/\log\mathcal{L}))$. It is hard as $840 \mid Q$, and $p > \ell_0 > T$ gives infinitely many distinct primes. Inverting: if $\mathcal{L} \ge (k+1)\log\log p$ then $W(p) > T \ge (\log p)^{k+1}$; otherwise $\mathcal{L}/\log\mathcal{L} \ll_k \log\log p/\log\log\log p$, and $\mathcal{L} \ge k\log\log p - O_k(\log\log p/\log\log\log p)$. All inputs are effective. □

For $k = 3$ there are no hyperedges, level 3 is empty, and the argument is Theorem 9.2 applied after the quarantine with $c_0 = \delta/32$ (its hypotheses (G), (W) follow from $w_\ell(\Pi) \le c_0$); this is the exponent-3 case with an absolute constant. The constants are astronomically lossy ($\delta = e^{-50}$, $\delta_r = [4er\rho^r]^{-1}$, and $A_k$ may be very large); all statements are asymptotic for fixed $k$, and we claim no rate for $k = k(T) \to \infty$.

## 13. Limits of the method

13.1. **Complete certificates.** A *complete certificate* for $W > T$ is a reduced class $c \bmod Q$ all of whose primes have $W(p) > T$. The class of one modulo $L^*(T) = \mathrm{lcm}(24,\ M \le T,\ M \equiv 3\ (4))$ is one, with $\log L^*(T) = (2/3 + o(1))T$.

**Proposition 13.1** (Proved)**.** *Every complete certificate has* $\log Q \ge (2/3 - o(1))T$.

*Proof.* (i) If a prime $\ell \equiv 3 \pmod 4$, $\ell \le T$, does not divide $Q$, then $c$ is compatible with the atom $-4 \in \mathcal{R}(\ell)$ ($D = 1$), and Dirichlet's theorem gives primes with $W \le \ell$; so $\prod_{\ell \le T, \ell \equiv 3(4)} \ell \mid Q$. (ii) Since $\mathcal{R}(3) = \{2\}$, $3 \mid Q$ and $c \equiv 1 \pmod 3$. (iii) Let $\ell \equiv 1 \pmod 4$, $\ell \le T/3$, and suppose a prime $r \equiv 2 \pmod 3$ divides $A = (3\ell+1)/4$. The atom $M = 3\ell$, $D = r$ has class $-4r \equiv 1 \pmod 3$; if $\ell \nmid Q$ it is compatible with $c$, a contradiction. (iv) The primes

$\ell \equiv 1 \pmod 4$ for which $(3\ell+1)/4$ has no prime factor $\equiv 2 \pmod 3$ avoid the classes $\ell \equiv -3^{-1} \pmod r$ for all primes $r \equiv 2 \pmod 3$, $r \geq 5$; by the prime number theorem in progressions their logarithmic weight up to $X$ is $o(X)$. Hence $\log Q \geq T/2 + T/6 - o(T)$. □

**Remark 13.2** (The previous record; proved modulo [2, Cor. 11])**.** Put $L^*(T) = \operatorname{lcm}(24,\ M \leq T,\ M \equiv 3\ (4))$. Every prime $p \equiv 1 \pmod{L^*(T)}$ has $W(p) > T$, by Lemmas 2.1 and 2.2. A prime $\ell \leq T/3$ divides some such $M$ (namely $\ell$ or $3\ell$). A prime $\ell \in (T/3, T]$ divides one iff $\ell \equiv 3 \pmod 4$. Higher prime powers contribute $O(\sqrt{T}\log T)$. Hence $\log L^*(T) = (2/3 + o(1))T$. Every prime factor $\ell$ of $L^*(T)$ has $\log \ell \leq \log T = o(\log L^*(T))$, so Chang's bound gives a prime $p \equiv 1 \pmod{L^*(T)}$ with $\log p \leq (12/5 + o(1)) \log L^*(T)$, i.e. $W(p) > T \geq (5/8 - \varepsilon)\log p$. As $840 \mid L^*(T)$ and $p > L^*(T) > T$, this holds for infinitely many hard primes.

Proposition 13.1 bounds $\log Q$, not $p$: a class $c \neq 1$ may contain primes below $Q$. What it shows is that a complete certificate whose prime is supplied by a least-prime theorem, which certifies $\log p$ only through an upper bound $O(\log Q)$, certifies no more than $W(p) \gg \log p$; with Chang's exponent, no more than $5/8$ by this route.

13.2. **Prime-local designs.** Call a design *prime-local* if it imposes $n \equiv 1 \pmod{\ell^{e_\ell}}$ for $\ell$ in a set $\Pi$ and every $M \leq T$, $M \equiv 3 \pmod 4$, has at most one prime factor outside $\Pi$. Theorem 6.1 uses such a design.

**Proposition 13.3** (Proved)**.** *In a prime-local design,* $\Pi$ *contains every odd prime* $\leq \sqrt{T/3}$ *with at most one exception. Hence every prime it produces has* $\log p \geq (1/\sqrt{3} - o(1))\sqrt{T}$.

*Proof.* For odd primes $\ell_1 < \ell_2 \leq \sqrt{T/3}$ outside $\Pi$, one of $\ell_1\ell_2, 3\ell_1\ell_2$ is $\equiv 3 \pmod 4$ and $\leq T$. □

13.3. **The hypothesis $H_{\min}$ and the hub obstruction.**

**Hypothesis 13.4** ($H_{\min}(\theta)$; proved for every $\theta > 0$ (Proposition 12.3))**.** For every $\varepsilon > 0$ and large $T$ there are $Q$ with $840 \mid Q$, $\log Q \leq T^{\theta+\varepsilon}$, and a minorant $B$ as in Theorem 5.1 with $\log \max d_i \leq T^{\theta+\varepsilon}$, $\log(M_1/\mu) \leq T^{\varepsilon}$, $\mu > 0$, and the twist condition.

**Theorem 13.5** (Proved modulo Theorem 4.1)**.** $H_{\min}(\theta)$ *implies* $W(p) > (\log p)^{1/\theta - \varepsilon}$ *for infinitely many hard primes, for every* $\varepsilon > 0$.

*Proof.* Adjoin to $Q$ a prime $\ell_0 \in (R, 2R]$, $R = \max(T, \max d_i)$; this keeps all hypotheses and forces $p > T$. Theorem 5.1 gives $\log p \leq T^{\theta + 2\varepsilon + o(1)}$. □

$H_{\min}$ is a purely combinatorial statement about congruences. Proposition 12.3 (via Theorem 11.5) establishes it for every $\theta > 0$, indeed with $T^{o(1)} = \exp(O_\theta(\log T/\log\log T))$. For $\theta < 1/2$ the residual atoms involve several free primes, and plain event-level Bonferroni truncation over the raw atom list fails:

**Proposition 13.6** (Proved)**.** *Fix $0<\theta<1/2$, put $y=T^{\theta}$, and impose the class of one modulo all prime powers $\le T$ of primes $\le y$. Take as events all atoms $(M,c)$, $c\in\mathcal{R}(M)$, compatible with this, and let $N(n)$ count those containing $n$. Then for large $T$ and every even $J$ with $(4.2\theta^2+o(1))(\log T)^2\le J\le y$, the Bonferroni minorant $B_J=\sum_{j<J}(-1)^j\binom{N}{j}$ has negative mean in the CRT measure on the class* 1.

*Proof.* The mean is $P(N=0)-E[\binom{N-1}{J-1}\mathbf{1}_{N\ge1}]$. Let $r=\lceil\sqrt{2J-1}\rceil\le\sqrt{2y}+1$, and take $r$ primes $\equiv1 \pmod 4$ and $r$ primes $\equiv3 \pmod 4$ in $(y,2y]$; there are enough of them. If $n\equiv-4$ modulo all of them, then for every cross pair $M=\ell\ell'\le4y^2\le T$ we have $M\equiv3 \pmod 4$, and $D=1$ gives the atom $-4$ mod $M$. So $N\ge r^2\ge2J-1$, on a set of measure $\ge(2y)^{-2r}$. Hence

$$E\tbinom{N-1}{J-1}\ge(2y)^{-2r}\tbinom{2J-2}{J-1}\ge\exp\big((J-1)\log4-\log(2J-1)-2r(\theta\log T+\log2)\big),$$

which exceeds $1>P(N=0)$ once $J>(2\sqrt{2}\theta/\log4)^2(\log T)^2(1+o(1))$. □

Proposition 13.6 is an obstruction only to truncation by the number of events. On the hub configuration of its proof, $r^2$ events fire on $2r$ primes; the error of the support truncation (Lemma 7.2) is at most $\binom{4r}{L}\le16^r$ there, against probability $(2y)^{-2r}$.

*Below $\theta=1/3$.* Support truncation alone is not enough once events involve three or more primes: Lemma 10.2 needs codegrees $\lesssim1/L$, and the small-height rational classes of Remark 11.6 violate this inside a single truncation. The per-prime conditions are supplied by the iterated quarantine (Lemma 12.2), and the codegree conditions by the level structure of Theorem 11.5, in which heavy sets are moved to lower levels whose truncation is independent of the codegree thresholds above them.

## 14. The slice parameter

For a prime $p$ let $\mathcal{B}_p$ be the set of pairs $(c,k)$ with $1\le k\le\lfloor2p/3\rfloor$, $1\le c\le\lfloor(2p+k)/(4k)\rfloor$, $(p,ck)=1$; put $h=4ck$, $N_{c,k}=p^2+4ck^2$, $s=\mathrm{sf}(c)$, and let $\chi_s=(\Delta_s/\cdot)$, where $\Delta_s<0$ is the discriminant of $\mathbb{Q}(\sqrt{-s})$; its conductor divides $4s\mid h$. $M_{c,k}(p)$ is the number of divisors of $N_{c,k}$ in the class $-p \pmod h$. The slice $(c,k)$ is *positive* if $p(a+b)=k(4abc-1)$ has a solution in positive integers $a,b$. This happens iff $M_{c,k}(p)>0$. Both directions rest on the identity, for $e=ha-p$, $f=hb-p$,

$$(ha-p)(hb-p)-N=h\big(4abck-p(a+b)\big)-4ck^2=h\big(k(4abc-1)-p(a+b)\big),$$

so $ef=N$ iff $k(4abc-1)=p(a+b)$. If $e\equiv-p \pmod h$ is a positive divisor of $N$, then $f=N/e\equiv p^2(-p)^{-1}\equiv-p \pmod h$ (as $\gcd(p,h)=1$), so $a=(p+e)/h$, $b=(p+f)/h$ are positive integers with $ef=N$. Conversely, if $k(4abc-1)=p(a+b)$ then $pa,pb<4abck$, so $e=ha-p>0$ and $f=hb-p>0$; and $ef=N$, $e\equiv-p \pmod h$.

**Lemma 14.1** (Genus obstruction; proved)**.** *If $p\equiv1 \pmod{24}$, $(c,k)\in\mathcal{B}_p$ and $\chi_s(p)=1$, then $M_{c,k}(p)=0$.*

*Proof.* Write $c = st^2$. Every prime $q \mid N$ is odd, prime to $stk$, and $(p(2tk)^{-1})^2 \equiv -s \pmod q$, so $\chi_s(q) = 1$; hence $\chi_s(e) = 1$ for every divisor $e$ of $N$. But $\chi_s(-p) = \chi_s(-1)\chi_s(p) = -1$, and $\chi_s$ is defined modulo $h$. □

**Lemma 14.2** (A sufficient prime factor; proved)**.** *If a prime $q$ divides $N_{c,k}$ and $q \equiv -p \pmod h$, then $M_{c,k}(p) \geq 1$.*

*Proof.* $q$ itself is a divisor in the class $-p$. □

**Lemma 14.3** (Proved)**.** *Let $p \equiv 1 \pmod{24}$ and suppose $(\ell/p) = 1$ for every prime $5 \leq \ell \leq B$. Then $ck_{\min}(p) > B$. In particular $ck_{\min}(p) \geq n_p$, the least quadratic non-residue modulo $p$.*

*Proof.* For $ck \leq B$, $s \leq B$ and $\chi_s(p) = (-s/p) = (-1/p)\prod_{q|s}(q/p) = 1$, since $(-1/p) = (2/p) = (3/p) = 1$. Apply Lemma 14.1. □

**Proposition 14.4** (Proved)**.** *Let $24 \mid L$ and let $a \bmod L$ be a reduced class with $a \equiv 1 \pmod{24}$ such that every sufficiently large prime $p \equiv a \pmod L$ has $ck_{\min}(p) > T$. Then every prime $5 \leq \ell \leq T$ divides $L$ and $(a/\ell) = 1$.*

*Proof.* Suppose $\ell \nmid L$ or $(a/\ell) = -1$. Choose $c_0 \bmod 4\ell$ with $c_0 \equiv a \pmod{\gcd(L, 4\ell)}$, $c_0 \equiv 1 \pmod 4$, $(c_0/\ell) = -1$; for $n \equiv 1 \pmod 4$, $\chi_\ell(n) = (n/\ell)$, so $\chi_\ell(c_0) = -1$. Take a prime $q \equiv -c_0 \pmod{4\ell}$, $q \nmid L$. Then $\chi_\ell(q) = \chi_\ell(-1)\chi_\ell(c_0) = 1$, so $-4\ell \equiv \rho^2 \pmod q$ for some $\rho \not\equiv 0$. The conditions $p \equiv a\ (L)$, $p \equiv c_0\ (4\ell)$, $p \equiv \rho\ (q)$ are compatible and define a reduced class, which contains infinitely many primes. For these, $q \mid p^2 + 4\ell$ and $q \equiv -p \pmod{4\ell}$, so $M_{\ell,1}(p) \geq 1$ (Lemma 14.2), and $(\ell, 1) \in \mathcal{B}_p$ for large $p$, whence $ck_{\min}(p) \leq \ell \leq T$. □

**Corollary 14.5** (Proved)**.** *Let $B(n)$ be a finite combination of residue classes with $B(p) \leq \mathbf{1}[ck_{\min}(p) > T]$ for all large primes $p$ in a reduced class $\equiv 1 \pmod{24}$. Then every large prime $p$ in that class with $B(p) > 0$ has $n_p > T$. Together with Lemma 14.3: the depth certifiable by congruences is exactly $n_p$.*

*Proof.* $B$ is periodic modulo some $L$, which we take divisible by the modulus of the given class (hence by 24). If $B(p) > 0$ then $B$ is positive on the whole class $p \bmod L$, which lies in the given class. So every large prime in the class $p \bmod L$ has $ck_{\min} > T$, and Proposition 14.4 gives $(p/\ell) = 1$, i.e. $(\ell/p) = 1$ by reciprocity ($p \equiv 1 \pmod 4$), for $5 \leq \ell \leq T$; also $(2/p) = (3/p) = 1$ as $p \equiv 1 \pmod{24}$. Hence $n_p > T$. □

**Theorem 14.6** (Proved modulo [8, Prop. 5.1])**.** $ck_{\min}(p) \gg \log p \cdot \log\log\log p$ *for infinitely many hard primes $p$.*

*Proof.* Lau and Wu [8, Prop. 5.1] (following Graham–Ringrose [5]; the numbering is that of the authors' preprint, and the published numbering was not compared) show the following. They phrase it via $\chi_p(q) = 1$ with $\chi_p = (p/\cdot)$, which for $p \equiv 1 \pmod 4$ equals $(q/p)$ by reciprocity, including $q = 2$. The statement: for small fixed $\delta > 0$ and $y(x) = \delta \log x \log_3 x$ there are $x_n \to \infty$

and primes $p \in (x_n^{1/2}, x_n \log x_n]$ with $p \equiv 1 \pmod 4$ and $(q/p) = 1$ for all primes $q \le y(x_n)$. Such $p$ are $\equiv 1 \pmod{24}$ and, since $(5/p) = (7/p) = 1$, lie in a Mordell class. Lemma 14.3 gives $ck_{\min}(p) > y(x_n) \ge (\delta/2) \log p \log_3 p$. Effectivity is not claimed. □

**Remark 14.7** (Attribution)**.** The Ω-bound in Theorem 14.6 is due to Graham and Ringrose [5]: $n_p = \Omega(\log p \log_3 p)$. The theorem transports it via the Yamamoto-type Lemma 14.3, using Lau–Wu's version only for the extra congruence $p \equiv 1 \pmod 4$. It is not a new bound for the least non-residue. What is new here is the equivalence in Corollary 14.5: congruence methods certify exactly $n_p$.

Under GRH, Montgomery's $n_p = \Omega(\log p \log\log p)$ would give the corresponding bound, and Ankeny's $n_p \ll (\log p)^2$ caps what congruences can certify; the random model predicts $\max_{p \le x} n_p \asymp \log x \log\log x$ (statements as quoted in [8]).

**Corollary 14.8** (Proved modulo Theorem 4.1)**.** *For every fixed $k \ge 3$ and infinitely many hard primes $p$, jointly $W(p) \ge (\log p)^{k-o(1)}$ and $ck_{\min}(p) \ge (\log p)^{1-o(1)}$.*

*Proof.* The primes $p$ of Theorem 1.1 (for this $k$) satisfy $p \equiv 1 \pmod Q$, where $24 \mid Q$ and every prime $\ell \le y = T^{1/k} \exp(2\mathcal{L}/\log \mathcal{L})$ divides $Q$ (Proposition 12.3). So $p \equiv 1 \pmod{24}$ and $(p/\ell) = 1$, hence $(\ell/p) = 1$ by reciprocity ($p \equiv 1 \bmod 4$), for every odd $\ell \le y$. Lemma 14.3 gives $ck_{\min}(p) > y$, and $\log p \le T^{1/k} \exp(O_k(\mathcal{L}/\log \mathcal{L})) = y \exp(O_k(\mathcal{L}/\log \mathcal{L}))$ with $\mathcal{L} \asymp_k \log\log p$ gives $y \ge (\log p)^{1-o(1)}$. □

## 15. The Haar side

Impose the class of one at all primes $\le z$ ($z \ge 3$), i.e. $n \equiv 1 \pmod{Q_z}$ with $Q_z$ defined like $Q$. For $M \le T$ write $M = mr$ with $m$ the $z$-smooth and $r$ the $z$-rough part. If $r > 1$, the surviving atoms of $M$ are the $D \mid A_M^2$ with $m \mid 4D + 1$, each giving the event $n \equiv -4D \pmod r$ of conditional Haar probability $1/\varphi(r)$.

**Lemma 15.1** (Proved)**.** *If $M \equiv 3 \pmod 4$, $D \mid A_M^2$ and $M = mr$ with $m \mid 4D + 1$, then $m \le r^2 + 1$.*

*Proof.* By the involution of Lemma 3.3 assume $D \le A$, $D = sr'^2$, $A = sr'k$, $k \ge r'$. As there, $m \mid r' + k \le 2k$, so $r = (4sr'k - 1)/m \ge 2sr' - 1/(2k)$, i.e. $r \ge 2sr'$, and $m \le 4sr'^2 + 1 \le r^2 + 1$. □

**Lemma 15.2** (Proved)**.** *For every $3 \le z \le T$, the total mass $S_{\text{tot}} = \sum 1/\varphi(r)$ over all surviving atoms is $\exp(O(\log T/\log\log T))$.*

*Proof.* Use $1/\varphi(r) \ll \log\log T \cdot m/M$ and repeat the proof of Lemma 3.3 with $\ell$ replaced by $r$; every surviving atom has $m \mid \gcd(4D + 1, M)$, hence $m \mid r' + k$. □

(Using [4, Prop. 1.4] on dyadic blocks one gets $S_{\mathrm{tot}} \ll (\log T)^4 \log\log T$; this is not needed below.)

**Theorem 15.3** (Warm-up; proved)**.** $\log(1/\delta^*(T)) \le T^{1/3+o(1)}$.

*Proof.* Fix $\varepsilon > 0$, put $y = T^{1/3+\varepsilon}$ and $z = y$. Every rough part is $\ell$, $\ell^2$ or $\ell_1\ell_2$ (as $y^3 > T$). Use the independent coordinates $X_\ell = n \bmod \ell^{e_\ell}$. Single atoms form forbidden sets $G_\ell$ of measure $g_\ell$, with $1 \notin G_\ell$ (Lemma 2.2) and $\sum g_\ell \le S_{\mathrm{tot}}$.

*Bad primes.* $B = \{\ell : \ g_\ell > 1/4\}$ has $|B| \le 4S_{\mathrm{tot}} = T^{o(1)}$. Impose also $n \equiv 1 \pmod{\ell^{e_\ell}}$ for $\ell \in B$ (cost $\le |B|\log T$). Pairs inside $B$ die by Lemma 2.2; a pair atom $(\ell b, a)$ with $b \in B$ becomes the residue $a$ mod $\ell$ at $\ell$ if $a \equiv 1$ $(b)$ and dies otherwise. The enlarged sets satisfy $g'_\ell \le 1/4 + |B|\tau_{\max}^2 T/(\ell y(\ell-1)) \le 1/2$, with $\tau_{\max}^2 = \max_{A\le T} \tau(A^2) = T^{o(1)}$, and $\sum(g'_\ell - g_\ell) \le |B|\tau_{\max}^2 T/y^2 = T^{1/3-2\varepsilon+o(1)}$.

*Good pairs.* Under the product measure $\mu'$ uniform on the allowed residues, a good pair event has $\mu'(E) \le 4/((\ell_1-1)(\ell_2-1))$, and the per-prime weight is $w'_\ell \le 16\tau_{\max}^2 T/(\ell^2 y) \le T^{-3\varepsilon+o(1)}$. The asymmetric local lemma [1, Lemma 5.1.1] applies to $\mu'$. Here $\mu'$ is the Haar measure conditioned on a product set, hence a product measure, and a pair event is mutually independent of all events sharing no prime with it. With $x_E = 2\mu'(E)$ we have $\sum_{E'\sim E} x_{E'} \le 2(w'_{\ell_1} + w'_{\ell_2}) = T^{-3\varepsilon+o(1)}$, so $\prod_{E'\sim E}(1-x_{E'}) \ge 1/2$ for large $T$. The lemma gives $\mu'(\text{no good pair event}) \ge \prod(1-x_E) \ge \exp(-16S_{\mathrm{tot}})$.

*Collecting.* With $Q' = Q_y \prod_{b\in B} b^{e_b}$,

$$\begin{aligned}\delta^*(T) &\ge \frac{8}{\varphi(Q')}\prod(1-g'_\ell)\, e^{-16S_{\mathrm{tot}}}\\ &\ge \exp\big(-\pi(y)\log T - |B|\log T - 18S_{\mathrm{tot}} - T^{1/3-2\varepsilon+o(1)}\big)\\ &= \exp\big(-T^{1/3+\varepsilon+o(1)}\big).\end{aligned}$$

□

The exponent $1/3$ reflects the crude bound $N(M) \le \tau(A^2)$ for multi-prime atoms at a fixed prime.

Let $w_\ell(T,z) = \sum 1/\varphi(r_E)$ over the distinct surviving events $E$ whose rough part is divisible by $\ell$.

**Lemma 15.4** (Proved)**.** *Let $3 \le z \le T$. If $w_\ell(T,z) \le \log z/(8\log T)$ for all primes $z < \ell \le T$, then $\delta^*(T) \ge (8/\varphi(Q_z))\exp(-4S_{\mathrm{tot}})$, so $\log(1/\delta^*(T)) \le \pi(z)\log T + 4S_{\mathrm{tot}} + O(1)$, with $S_{\mathrm{tot}}$ as in Lemma 15.2.*

*Proof.* Apply the local lemma [1, Lemma 5.1.1] on the coordinates $X_\ell$, $\ell > z$ (uniform on units), to all distinct surviving events, with $x_E = 2/\varphi(r_E) \le 1/2$ and $E \sim E'$ when the rough parts share a prime. An event has at most $\log T/\log z$ primes, so $\prod_{E'\sim E}(1-x_{E'}) \ge \exp(-4\sum_{\ell\mid r_E} w_\ell) \ge e^{-1/2} \ge 1/2$, and the lemma gives probability $\ge \prod(1-x_E) \ge \exp(-4S_{\mathrm{tot}})$ of no event. The class of one modulo $Q_z$ has relative measure $\varphi(24)/\varphi(Q_z)$, and $\log\varphi(Q_z) \le \pi(z)\log T$. □

**Remark 15.5** (Conditional; the implication is proved modulo [4, Prop. 1.4])**.** Suppose the hypothesis of Lemma 15.4 holds with $z = (\log T)^C$ for all large $T$. Then $\log(1/\delta^*(T)) \ll (\log T)^{\max(C+1,4)+o(1)}$, since $S_{\mathrm{tot}} \ll (\log T)^4 \log\log T$ by [4, Prop. 1.4]. Without [4], Lemma 15.2 gives only $\delta^*(T) \geq \exp(-T^{o(1)})$ by this argument. We do not know how to prove the per-prime hypothesis.

For $y = T^{1/3+o(1)}$ the Haar bound of Theorem 15.3 has a prime-side counterpart: the case $k = 3$ of Theorem 1.1, through the pointwise minorant of Theorem 9.2. The local lemma by itself does not provide a pointwise minorant; the multilevel minorants of §11 provide one for every $y = T^\theta$.

**Single-prime atoms and the Type I variety.** Following [4, §2], let $\Sigma_I^n$ be the set of sextuples $(a, b, c, d, e, f)$ satisfying their identities (2.1)–(2.9), among them

$$4abd = ne+1, \quad ce = a+b, \quad 4acd = n+f, \quad ef = 4a^2d+1, \quad bf = na+c,$$

and put $F_I(n) = \#\{(a, b, c, d, e, f) \in \mathbb{N}^6 \cap \Sigma_I^n : \ a \leq b\}$.

**Lemma 15.6** (Proved)**.** *Let $r > 1$ be odd. The atoms $(M, D)$ with $M = mr \equiv 3 \pmod 4$, $D \mid A_M^2$, $m \mid 4D+1$ and $D \leq A_M$ (any $m \geq 1$) correspond bijectively to the points of $\mathbb{N}^6 \cap \Sigma_I^r$ with $a \leq b$ and $d$ squarefree, via $D = sr'^2$, $A_M = sr'k$ ($s$ squarefree) and*

$$(a, b, c, d, e, f) = (r',\ k,\ (r'+k)/m,\ s,\ m,\ (4D+1)/m).$$

*The class of the atom is $-4D \equiv -a/b \pmod r$. Hence the number of distinct classes modulo $r$ carried by atoms with rough part $r$, under any class-of-one quarantine, is at most $2F_I(r)$.*

*Proof.* By the proof of Lemma 3.3, $m \mid r'+k$, so $c \in \mathbb{N}$. Then $4abd = 4A_M = mr+1$, $ce = a+b$ and $ef = 4sr'^2+1$; by direct computation these three identities (with nonzero entries) imply the others, e.g. $e \cdot 4acd = 4a^2d + 4abd = (ef-1) + (ne+1)$ and $e \cdot bf = a(ne+1) + b = e(an+c)$. Conversely a point with $a \leq b$ and $d$ squarefree gives $D = da^2 \leq A = dab$, $M = ne = 4A-1$, $m = e \mid 4D+1$. The triple $(s, r', k)$ determines $D$ and $A$, hence $M$; and $4A \equiv 1 \pmod r$ gives $-4D \equiv -r'/k$. Partners $A^2/D$ at most double the count. □

So the single-prime forbidden sets are governed by the Type I variety $\Sigma_I^r$. Modulo the proof of [4, Prop. 1.7] (its Lemma 2.8 and §3), which bounds the relevant points, $F_I(n) \leq n^{3/5+O(1/\log\log n)}$; ET remark that 3/5 appears to be the limit of divisor-bound arguments. The following elementary bound is not used below.

**Proposition 15.7** (Proved)**.** *For a prime $\ell$, the points of $\mathbb{N}^6 \cap \Sigma_I^\ell$ with $a = 1$ number $\ll \sqrt{\ell}\log\ell$.*

*Proof.* A point is determined by $(a, c, f)$, via $4acd = \ell + f$, $ef = 4a^2d+1$, $b = ce - a$. First $\ell \nmid c$: otherwise $\ell$ divides $y = acd$, $z = bcd$ and $x = abd\ell$ in $4/\ell = 1/x + 1/y + 1/z$, giving $4 \leq 3$. So $g = \gcd(c, f)$, which divides

$\ell a + c$ (as $f \mid \ell a + c$ by $bf = \ell a + c$) and $\ell + f$ (as $c \mid \ell + f$), divides $\ell$ and hence equals 1. Then $f \mid a\ell + c$ and $c \mid a(\ell + f)$ give $cf \mid a(\ell + f) + c$, so $(c - a)(f - 1) \le a(\ell + 1)$. With $a = 1$: either $f \le 1 + \sqrt{\ell + 1}$, and then $c \mid \ell + f$; or $c \le 1 + \sqrt{\ell + 1}$, and then $f \mid \ell + c$. So the count is at most $2\sum_{j \le 2+\sqrt{\ell}} \tau(\ell + j)$, and $\tau(n) \le 2\#\{\delta \mid n : \delta \le \sqrt{n}\}$ gives $\sum_{j \le 2+\sqrt{\ell}} \tau(\ell + j) \le 2\sum_{\delta \le \sqrt{2\ell}} (3 + \sqrt{\ell})/\delta + O(\sqrt{\ell}) \ll \sqrt{\ell} \log \ell$. □

**Theorem 15.8** (Proved implication)**.** *Suppose $F_I(n) \le n^{\eta + o(1)}$ for all $n$, for some $0 \le \eta \le 1$, and put $\kappa = \eta/(1 + \eta)$. Then $w_\ell(T, z) \le T^{\kappa + o(1)}/\ell$ for all $z$ and all primes $\ell \le T$, and $\log(1/\delta^*(T)) \le T^{\kappa + o(1)}$.*

*Proof.* By Lemma 15.6 and the crude count, the number of events with rough part $r$ is at most $\min(2F_I(r), \tau^{*2}T/r)$, with $\tau^*$ as in §12. Hence

$$w_\ell \ll \log\log T \sum_{r \le T,\ \ell \mid r} \frac{\min(2r^{\eta + o(1)}, \tau^{*2}T/r)}{r}.$$

Splitting at $R = T^{1/(1+\eta)}$ gives $w_\ell \ll R^\eta T^{o(1)}/\ell + \tau^{*2}T/(\ell R) = T^{\kappa + o(1)}/\ell$. With $z = T^{\kappa + \varepsilon}$ the hypothesis of Lemma 15.4 holds for large $T$, and Lemma 15.4 with $S_{\mathrm{tot}} = T^{o(1)}$ (Lemma 15.2) gives $\log(1/\delta^*) \le \pi(z) \log T + T^{o(1)}$. □

With $\eta = 3/5$ this gives $\kappa = 3/8$, weaker than Theorem 15.3; $\eta = 1/2$ gives exactly $1/3$. So individual Type I bounds beat the exponent $1/3$ only if $\eta < 1/2$, well beyond [4, Prop. 1.7]. The single-prime part of the per-prime hypothesis is not the bottleneck (Theorem 15.3 absorbs it by quarantining $T^{o(1)}$ bad primes); what is needed is control of $\sum_{r \le T,\ \ell \mid r} F_I(r)/r$ uniformly in $\ell$, a progression average of Type I counts (Assessment). The next theorem makes this route unnecessary.

**Iterated quarantine.** For an atom $(M, D)$ and a set $\Pi$ of primes let $m_\Pi$ be the $\Pi$-part of $M$ and $r_\Pi = M/m_\Pi$; the atom *survives* $\Pi$ if $m_\Pi \mid 4D + 1$ and $r_\Pi > 1$. Put $S^* = \sum_{(M,D)} \max_\Pi \mathbf{1}[\text{survives } \Pi]/\varphi(r_\Pi)$.

**Theorem 15.9** (Proved; the polylogarithmic form modulo [4, Prop. 1.4])**.**

$$\log(1/\delta^*(T)) \ll (\log T)^3 (S^* + 1).$$

*Hence* $\log(1/\delta^*(T)) \le \exp(O(\log T/\log\log T)) = T^{o(1)}$*, and, modulo* [4, Prop. 1.4]*,* $\log(1/\delta^*(T)) \ll (\log T)^7/\log\log T$.

*Proof.* Lemma 12.1 gives $S^* \le \exp(O(\log T/\log\log T))$, and its proof with [4, Prop. 1.4] gives $S^* \ll (\log T)^4 \log\log T$. Let $z = (\log T)^2$, $k = \lfloor \log T/\log z \rfloor$, $c_0 = 1/(8k)$, and apply Lemma 12.2: at most $kS^*/c_0$ primes are added, and at the end every free prime has $w_\ell \le c_0$ and every event has at most $k$ free primes, and the local lemma with $x_E = 2P(E)$ applies ($\sum_{E' \sim E} x_{E'} \le 2kc_0 = 1/4$), giving probability $\ge \exp(-4S^*)$ of no event. The class of one modulo the quarantined prime powers costs $\le (\pi(z) + 8k^2 S^*) \log T$. □

This improves Theorem 15.3, and it makes the per-prime hypothesis of Lemma 15.4 and of Remark 15.5 unnecessary. On the prime side the same quarantine (Lemma 12.2) supplies the per-prime conditions of Theorems 9.2 and 11.5, at the cost of $T^{o(1)}$ further quarantined primes, and the levels of Theorem 11.5 handle the codegrees, so both sides reach $T^{o(1)}$. For $T \le 10^6$ the iteration adds a few hundred primes in a single round, and the resulting certificates give $\log(1/\delta^*(10^6)) \le 3.6 \cdot 10^3$ (EVIDENCE).

## 16. Summary of status

- Proved: Lemmas 2.1–3.3, 7.1–8.2, 10.1, 10.2, 11.1, 11.3, 12.1, 12.2, 14.1–14.3, 15.1, 15.2, 15.4, 15.6; Lemma 9.1 (classical); Theorems 9.2, 11.2, 11.5; Corollary 11.4; Proposition 12.3, i.e. $H_{\min}(\theta)$ for every $\theta > 0$; Propositions 13.1, 13.3, 13.6, 14.4; Corollary 14.5; Theorems 15.3 and 1.3 (with the classical local lemma and divisor bound); the lower bound in Remark 11.6 (with the prime number theorem in progressions).
- Proved modulo Thorner–Zaman [15] (with the McCurley statement quoted there): Lemma 4.2, Theorems 5.1, 1.1, 6.1, 13.5, Corollary 14.8; hence the falsity of "$W(p) \le (\log p)^A$ for all large $p$" for every $A$, and $\log L_h(T) \le T^{o(1)}$ for the least hard prime $L_h(T)$ with $W > T$.
- Proved modulo Lau–Wu [8, Prop. 5.1]: Theorem 14.6.
- Proved modulo Chang [2, Cor. 11]: the 5/8 record (Remark 13.2).
- Proved: Theorem 15.9 ($\log(1/\delta^*) \le T^{o(1)}$); its polylogarithmic form modulo [4, Prop. 1.4].
- Proved modulo the proof of [4, Prop. 1.7]: $F_I(n) \le n^{3/5+o(1)}$. Proved implication: Theorem 15.8.
- Conditional on the per-prime hypothesis, and proved modulo [4, Prop. 1.4]: the polylogarithmic Haar bound of Remark 15.5.
- Open: an explicit rate for $W(p)$ beyond every fixed power of $\log p$ (e.g. $W(p) \ge \exp((\log\log p)^{1+\delta})$ infinitely often); an unconditional polylogarithmic Haar bound; $ck_{\min}(p) > (\log p)^{1+\delta}$ infinitely often.

Nothing here bears on the truth of the Erdős–Straus conjecture: a prime with large $W(p)$ or $ck_{\min}(p)$ may well have other representations.

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